Medals and Mindsets: How Women's Olympic Competitiveness Advances Gender Equality*

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Abstract

Can female athletic success reshape beliefs about women's competitive capacity and advance gender equality? We study this question in the context of China's return to the Olympics, combining comprehensive athlete records from 1984–2004 with census data. Exploiting variation in the timing of each prefecture's first female Olympic medalist and differential exposure across birth cohorts, we find that exposure to hometown female medalists narrows gender gaps in education, while male medalists produce no comparable effects. To address potential endogeneity, we employ a shift-share instrument leveraging changes in Soviet presence (shaped by boycott and dissolution) as an exogenous shock to medal opportunities across sports. We further show that media coverage framing female achievements as evidence of women's competitiveness helps explain these effects, and that female medalists shift fathers' beliefs about daughters' competitive capacity rather than directly inspiring girls. Our findings suggest that publicized female success can challenge gender stereotypes and promote equality.

Keywords: Competitiveness, Gender equality, Olympic Games, Media

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1. Introduction

The belief that women are less capable in competitive environments continues to reinforce gender inequality across education, labor markets, and household decision-making (Niederle and Vesterlund 2007; Buser, Niederle, and Oosterbeek 2014; Flory, Leibbrandt, and List 2015; Blau and Kahn 2017; Bursztyn, Fujiwara, and Pallais 2017; Bordalo, Coffman, Gennaioli, and Shleifer 2019; Carlana 2019). Sporting events featuring female athletes offer a rare opportunity to challenge this stereotype by displaying women's strength, discipline, and ability to excel in competition. However, little is known about whether—and through what channels—female athletic success can reshape beliefs about women's competitiveness and promote broader gender equality.

In this study, we focus on the Olympic Games—the most influential global arena for athletic competition. The Olympics bring together elite athletes from more than 200 countries, competing for medals before a worldwide audience of billions. This unparalleled visibility generates intensive media coverage and public attention, creating a unique setting to observe how female athletic success shapes social perceptions. Since their inception, the Games have undergone a profound transformation: from the complete exclusion of women in 1896 to near gender parity by 2024, with female athletes now competing and winning medals across nearly all disciplines.

We examine whether female Olympic success affects gender inequality in education. We focus on education for two main reasons. First, educational attainment is closely tied to beliefs about competitiveness. A large body of evidence shows that perceptions of women as less competitive underlie gender disparities in educational and occupational choices (Niederle and Vesterlund 2011; Buser, Niederle, and Oosterbeek 2014). Second, parents play a central role in shaping children's educational investments, and their gendered perceptions of competitiveness contribute to educational inequality (Tungodden and Willén 2023). The success of female Olympic athletes may shift these beliefs by providing highly visible and credible evidence of women's ability to excel in elite competition, thereby narrowing gender gaps in educational attainment.

We investigate this hypothesis in the context of China's return to the Olympic Games in 1984, a setting that offers several distinct advantages. Unlike countries with continuous participation–where the growth of women's sports and progress in gender equality evolved together–China had been absent from the Olympics for more than three decades due to Cold War tensions, and therefore had no influence on the Olympic movement itself.[1] China's reintegration in 1984 thus introduced a sudden and exogenous exposure

[1]For example, Title IX of the U.S. Education Amendments of 1972 exemplifies this entanglement: it simultaneously expanded female athletic participation and reflected growing commitment to gender equality.

to international women's sports for domestic audiences. At the time, Chinese society exhibited deep-rooted gender bias, particularly in education. Moreover, the Chinese education system functions as a multi-tiered tournament, in which advancement to elite secondary schools and universities depends on intense competition in entrance examinations. In this context, parental beliefs about children's competitiveness are especially consequential for educational outcomes.

While Olympic victories spark national celebration, their influence is especially strong in athletes' hometowns, where local pride, media coverage, and community identification amplify exposure to female competitive success.[2] We construct a dataset linking Chinese Olympic medalists to gender-related outcomes in their hometowns. The dataset includes all Chinese Olympic athletes from 1984 to 2004–the period beginning with China's return to the Games and ending just before the 2008 Beijing Olympics. For each athlete, we record gender, sport, competition year, medal type, and hometown prefecture. We then link these data to the 2015 Population Mini-Census, which reports detailed information for over one million individuals, including educational attainment and birth prefecture. This linkage allows us to examine how exposure to female Olympic success affected gender disparities across regions and cohorts.

Our empirical strategy exploits two complementary sources of variation. First, we use within-prefecture variation across cohorts, comparing individuals who were still making educational decisions with those who had completed schooling when their hometown produced its first female Olympic medalist. Specifically, we define the *Medal Cohort* as individuals who were age 18 or younger when their prefecture first produced a female medalist. The age 18 cutoff reflects that individuals below this threshold are typically still making schooling choices and thus more responsive to external influences, whereas older cohorts have largely completed their education. We verify the robustness of our results using alternative cutoffs (ages 12 and 15) and a continuous measure of exposure duration. This structure allows us to compare individuals from the same prefecture who experienced local female Olympic success at different points in their educational trajectories.

Second, we exploit variation in the timing of each prefecture's first female Olympic medal, which differed widely across regions between 1984 and 2004. Our sample includes 91 prefecture-level cities that had produced at least one female medalist by 2004, providing substantial regional variation in the timing of first successes. Using this staggered timing, we implement a fixed-effects model to estimate the impact of female Olympic success on gender disparities in educational attainment and related outcomes.

Our fixed-effects estimates document a strong association between hometown female

[2] We document several patterns consistent with this localized salience, reflected in both gazetteer commemorations and local internet search interest (Section 2.2)

Olympic success and a narrowing of the educational gender gap. Exposure to a female medalist during the educational decision window is associated with a 0.2-year increase in women's schooling relative to men—closing roughly one-fifth of the 0.99-year baseline gap observed among cohorts born before China's 1984 Olympic return. The association is specific to female athletes: an analogous measure for male medalists is uncorrelated with gender outcomes. This asymmetry mirrors the stereotype itself, which questions women's—not men's—competitive capacity; only visible female success carries information capable of revising the prevailing belief. The same null also serves as a placebo, suggesting that the pattern reflects female athletic achievement rather than general Olympic success or contemporaneous regional development that would accompany any medal-winning prefecture.

This pattern is further sharpened by heterogeneity across sport types. We classify sports along two dimensions. The first contrasts head-to-head sports (e.g., basketball), in which competitive prowess is on direct display, with performance-based sports (e.g., diving), which are judged against fixed standards. The second contrasts strength-oriented disciplines (e.g., weightlifting), which rebut the belief that women are physically weaker, with skill-oriented ones (e.g., rhythmic gymnastics), which are more readily reconciled with stereotypes of female grace. The female-medalist effect is significantly larger for head-to-head and strength-oriented sports and smaller for the others—consistent with the interpretation that the main effect is concentrated in sports that most directly contradict gender stereotypes about women's competitive capacity.

To further trace where the narrowing of educational gender gap originates, we decompose educational attainment into two margins: *transitions*—advancement to the next school tier conditional on completing the previous one—and *dropouts*—exit before completion conditional on enrollment. The two margins capture distinct family decisions: transitions involve whether to advance a child to the next tier, whereas dropouts involve withdrawing an already-enrolled child, typically in response to financial shocks or other household difficulties. The gains operate almost entirely through the transition margin and are concentrated at the junior- and senior-high transitions, where China's selective tracking intensifies and parental decisions weigh most heavily. Dropout rates are essentially unchanged at every stage.

The main identification concern is that the emergence of female medalists may not be fully exogenous. One potential threat arises from time-varying factors that simultaneously influence a region's likelihood of producing female Olympians and its local gender-equality outcomes. Another is reverse causality: regions with improving gender equality may actively promote female participation in competitive sports, thereby increasing the likelihood of athletic success and confounding our interpretation.

To address these identification concerns, we construct an instrumental variable at the prefecture–Olympic-year level that captures exogenous variation in each prefecture's potential to win female Olympic medals. The instrument follows a shift-share design, combining pre-determined prefecture characteristics with external shocks generated by sharp changes in Soviet athletes' Olympic presence. Through a centralized, state-sponsored athletic system, the Soviet Union dominated women's Olympic events during much of the Cold War era. Its boycott of the 1984 Los Angeles Games and subsequent dissolution in 1991 sharply reduced Soviet participation, creating a competitive vacuum that substantially altered medal prospects, particularly for China. The resulting shift varied across sports: disciplines formerly dominated by Soviet athletes experienced greater new opportunities, while those with weaker Soviet presence changed little. Consequently, the magnitude of this exogenous shock differed across prefectures according to their pre-existing strengths across sports.

To operationalize this approach, we measure each prefecture's pre-existing sports specialization using medalist data from China's 1983 National Games–held just before the country's full Olympic reintegration. We then interact these fixed local strengths with time-varying Soviet female participation rates by sport to generate predicted medal opportunities for each prefecture in each Olympic year. The instrument therefore captures how the Soviet Union's collapse reshaped China's medal prospects across prefectures, depending on their historical composition of sports strengths.

The key identifying assumption in our shift-share IV strategy is that changes in Soviet participation rates were exogenous to gender-equality trends across Chinese prefectures (i.e., "shift-view"). We interpret the Soviet collapse as an external shock that affected China's medal opportunities but was unrelated to local social or cultural dynamics. We assess this assumption using several validation exercises–including balance checks, pre-trend analyses, and robustness tests with alternative specifications–all of which consistently support our identification strategy.

Our IV estimates align closely with the baseline results, showing statistically significant and economically meaningful effects of female medalists in reducing gender gaps in education. Exposure to female Olympic success increases women's schooling by about 0.56 years relative to men, consistent with the fixed-effects estimates in direction and statistical significance. Applying analogous instruments for male medal exposure yields no significant effects on gender-equality outcomes, reinforcing that the observed impacts stem specifically from female Olympic achievement.

Having established a causal link between female Olympic success and reduced gender inequality, we next examine the mechanisms driving this relationship. We posit that such achievements challenge stereotypes about women's competitiveness by providing highly visible and credible evidence of success in the most demanding arenas. Two main channels may underlie this effect. First, media coverage amplifies female athletes' accomplishments and disseminates new information about women's competitive ability to broad audiences, potentially reshaping social beliefs about gender and competition. Second, exposure to female success may directly influence parental perceptions of their daughters' competitiveness, prompting greater educational investment in girls. We investigate these mechanisms through analyses of media coverage patterns of medalists and survey data on parental beliefs and educational investments.

To examine the media channel, we analyze how Olympic medalists were portrayed in *People's Daily*, China's most authoritative newspaper during our study period. We quantify coverage along two dimensions: reporting intensity and thematic framing. Specifically, we distinguish between *competitive-equality* themes, which highlight women's competitive capabilities and equal potential for success, and *national-pride* themes, which emphasize collective achievement and national glory. The analysis shows that male and female medalists receive comparable overall coverage, but female athletes are significantly more likely to be framed through a competitive-equality lens. Moreover, among female medalists, stronger competitive-equality framing produces larger improvements in local gender outcomes than greater coverage intensity alone, indicating that the interpretation of female success matters more than its visibility.

We next examine whether exposure to female Olympic success shifts parental beliefs and household investment decisions. Using representative household surveys–the China Family Panel Studies (CFPS) and the China Household Income Project (CHIP)—we document two main patterns. First, fathers exposed to hometown female medalists hold higher educational expectations for daughters relative to sons, with no comparable effects observed for mothers. This asymmetry suggests that female Olympic success primarily challenges fathers' stereotypes about daughters' competitiveness, consistent with evidence that paternal biases shape gender-differentiated investments (Yi, Heckman, Zhang, and Conti 2015; Bhaskar, Li, and Yi 2023; Tungodden and Willén 2023). Second, these shifts in belief translate into changes in household spending: hometown female medalists reduce gender gaps in education-related expenditures, as parents increase tuition and training spending on daughters relative to sons while keeping non-educational spending unchanged. Notably, we find no direct effects on girls' own self-perceptions or aspirations, including their educational expectations, self-assessed competence, or career goals.

Our paper contributes to several strands of economic research. First, we add to the literature on gender differences in competitiveness and their role in sustaining gender inequality. A substantial body of work shows that women are systematically less likely than men to enter and persist in competitive settings, helping explain persistent gender gaps in educational choices, career outcomes, and earnings (Niederle and Vesterlund 2007; Niederle and Vesterlund 2011; Buser, Niederle, and Oosterbeek 2014; Reuben, Wiswall, and Zafar 2017; Buser, Niederle, and Oosterbeek 2024).[3] Growing evidence highlights that these differences are shaped by stereotypes and beliefs held by individuals, parents, teachers, and the wider social environment (Carlana 2019; Bordalo, Coffman, Gennaioli, and Shleifer 2019). Parental beliefs appear especially influential: Tungodden and Willén (2023) show that parents hold gender-differentiated beliefs about their children's competitiveness, which substantially affect educational choices, particularly for daughters. In China, for example, daughters are substantially less likely than sons to retake the *Gaokao* after initial failure (Kang, Lei, Song, and Zhang 2024).

We build on this literature by showing that elite women's sports can *change* parents' perceptions of their daughters' competitiveness and potential–demonstrating that such beliefs are malleable rather than fixed–and that media coverage plays a critical role in transmitting these belief changes.

Second, we also contribute to the growing literature on the socioeconomic impacts of global sporting events. Prior research has primarily examined the economic consequences of the Olympic Games–such as growth, employment, infrastructure, and trade (Bernard and Busse 2004; Rose and Spiegel 2011; Brückner and Pappa 2015)—although Baade and Matheson (2016) find that these effects are typically modest and short-lived. Other studies document temporary improvements in well-being or environmental quality (Kavetsos and Szymanski 2010; Chen, Jin, Kumar, and Shi 2013; Dolan, Kavetsos, Krekel, Mavridis, Metcalfe, Senik, Szymanski, and Ziebarth 2019), and more recent evidence shows that national football victories can build social cohesion and reduce ethnic divisions (Depetris-Chauvin, Durante, and Campante 2020). We enrich this literature by identifying persistent, non-economic effects of Olympic success: demonstrating that sporting achievements can reshape social beliefs and narrow gender inequality–core dimensions of long-term societal change.

Third, we contribute to the literature on the media's role in shaping social attitudes by examining not only what content is presented but how it is framed. Prior research shows that media exposure can substantially influence gender-related outcomes: television improved women's social status in India (Jensen and Oster 2009), soap operas reduced fertility in Brazil (La Ferrara, Chong, and Duryea 2012), reality television lowered teen pregnancy rates in the United States (Kearney and Levine 2015), and information campaigns increased female labor-force participation in Saudi Arabia (Bursztyn,

[3]Niederle and Vesterlund (2007) demonstrate in laboratory experiments that women are less likely to choose tournament-style compensation even when they perform as well as men. Buser, Niederle, and Oosterbeek (2014) show that competitive preferences predict gender differences in high school track choices and subsequent educational outcomes.

González, and Yanagizawa-Drott 2020).[4] We advance this literature by showing that *how* athletic achievements are framed–specifically, whether female victories emphasize competitive capability and excellence–has greater influence on gender attitudes and outcomes than the sheer intensity of media coverage.[5]

Finally, our findings do not rule out role-model dynamics in general but point more directly to a broader social channel of gender-attitude change—one that does not appear to operate through same-gender identification. The role-model literature shows that exposure to successful individuals sharing similar characteristics can reshape beliefs and influence behavior (Dee 2004; Carrell, Page, and West 2010; Beaman, Duflo, Pande, and Topalova 2012; Porter and Serra 2020).[6] Role-model effects typically operate through individual-level identification, often involving direct contact or targeted interventions within specific settings. In a parallel study, Guo, Ling, and Wen (2025) finds that female Olympic success reduces preference for boys in fertility decisions in China using two waves of provincial survey data before and after the 2016 Olympics, interpreting the effect as role-model driven. However, our evidence indicates broader social transmission: widely publicized displays of female competitive success influence parental beliefs and investment decisions that shape girls' opportunities.[7]

2. Institutional Background

2.1. Gender Equality in the Olympic Games

The modern Olympic Games constitute the world's largest international sporting event, bringing together more than 11,000 athletes from over 200 nations to compete across 33 sports. Beyond athletic performance, the Olympics serve as a global cultural platform watched by more than 3 billion viewers worldwide and heavily covered by international media. Their unparalleled reach and visibility make them a potential arena

[4]More broadly, media exposure has been shown to shape attitudes and behaviors across diverse contexts, including political participation (Gentzkow 2006; Enikolopov, Petrova, and Zhuravskaya 2011; Yanagizawa-Drott 2014; Adena, Enikolopov, Petrova, Santarosa, and Zhuravskaya 2015), social capital formation (Olken 2009), consumption choices (Bursztyn and Cantoni 2016), and violent behavior (Dahl and DellaVigna 2009). DellaVigna and Gentzkow (2010) provide a comprehensive review.

[5]The importance of shifting beliefs is also highlighted by Caria, Crepon, Krafft, and Nagy (2025), who show that interventions providing information and reducing practical barriers had limited effects on female employment in Egypt when underlying gender norms remained unchanged.

[6]The literature identifies several mechanisms through which role models influence behavior and outcomes: providing information about previously unknown opportunities and returns (Jensen 2010; Nguyen 2008); updating beliefs about one's own ability to succeed in those fields (Beaman, Duflo, Pande, and Topalova 2012); and disrupting stereotypes by demonstrating that success is possible for underrepresented groups (Carrell, Page, and West 2010; Porter and Serra 2020).

[7]Several patterns in our results are not consistent with a same-gender role-model channel, in which observed individuals primarily update through identification with successful athletes of their own gender: exposure does not shift girls' own self-perceptions or realized sport-related career choices, and male medalists generate no comparable educational effects for boys.

for transmitting values and beliefs, influencing how societies perceive competition, excellence, and gender roles.

The early Olympic Games reflected entrenched gender stereotypes that constrained women's participation. Women were excluded at the inaugural 1896 Games, and their share of competitors rose only gradually over the following six decades.[8] Gender parity emerged only later, driven by Western policy reforms—most prominently Title IX of the U.S. Education Amendments of 1972—and the IOC's institutional response (Brake 2010), which roughly tripled the number of women's events between 1972 and 2000.[9]

China was not a driver of this transformation but was instead shaped by it. After more than three decades outside the Olympic movement, China's return in 1984 brought Chinese audiences not just a major sporting event but an exogenous cultural shock—one that introduced a highly visible platform for showcasing women's competitive excellence. This distinct trajectory makes China a compelling setting for studying how female Olympic success influences gender equality.

2.2. China and the Olympic Games

China withdrew from the IOC in 1958 over disputes regarding Taiwan's representation, having last competed at the 1952 Helsinki Games. After Olympic membership was restored in 1979, China's participation in the 1984 Los Angeles Games marked a renewed engagement with the international community after a 32-year absence and a deliberate effort to project modernization through sport (Hong 2013). The Games quickly became a symbol of national progress and a vehicle for instilling national pride, transforming an imported sporting institution into a central element of Chinese national identity.

China's athletic system adopted the Soviet model of state-sponsored sports development, combining centralized planning with decentralized provincial implementation. Provincial and municipal governments invested in sports development to compete for national recognition, specializing in different disciplines based on local infrastructure and geographic advantages (Li, Meng, and Wang 2009).[10] Athletes excelling in provincial competitions, particularly the National Games, were selected for the national team to represent China internationally.

[8] Although 22 women competed across five sports in 1900, the International Olympic Committee (IOC) emphasized "grace and beauty" over strength or endurance, admitting tennis (in 1900) and figure skating (in 1908) but excluding track and field until 1928. Female participation rose from 2.2% in 1900 to 13% by 1960.

[9] Female participation rose from 23% at the 1984 Los Angeles Games to approach parity by 2024, with women entering events once viewed as beyond their physical capacity, including the marathon (added in 1984), weightlifting (2000), and boxing (2012).

[10] For example, coastal provinces such as Guangdong developed strong diving programs, Hubei and Hunan focused on gymnastics and badminton, while northern provinces like Heilongjiang specialized in winter sports.

Olympic athletes routinely become national celebrities, with their life stories, training journeys, and personal philosophies attracting broad public attention (Lu 2011). Medalists receive extensive coverage across state and local media, where their achievements are framed as expressions of national pride and personal excellence.

Olympic medalists attract exceptional attention and recognition in their home regions. Local newspapers, broadcasters, and community institutions provide intensive and sustained coverage of medalists' training, competitions, and post-Olympic lives, sustaining their visibility long after the Games conclude. Their prominence extends even into official historical records: 82% of Chinese Olympic medalists are commemorated in their home county's gazetteer—official local chronicles that document notable figures and events as enduring records of community heritage and pride—with an average of three entries per medalist.[11]

2.3. The Soviet Collapse and China's Olympic Opportunity

The dissolution of the Soviet Union in December 1991 marked one of the most consequential geopolitical shifts of the late twentieth century, transforming global politics and reshaping international sport. The collapse dismantled the world's most successful state-sponsored athletic system, opening unprecedented space for new Olympic powers–most notably China.

Before 1991, the Soviet Union dominated Olympic competition through a centralized sports system established in the 1920s to serve both physical and ideological goals (Parks 2016). Designed to showcase socialist superiority, this model emphasized early talent identification, full-time professional training, and heavy state investment in scientific support and facilities. Between 1952 and 1988, Soviet athletes topped the Olympic medal tables at six of ten Games, excelling across major sports.

The Cold War further politicized Olympic contests. Following the Soviet invasion of Afghanistan, the U.S. led a boycott of the 1980 Moscow Olympics, and the Soviet Union reciprocated at Los Angeles in 1984. These actions demonstrated how Olympic performance had become an extension of global rivalry.

Soviet dissolution abruptly fractured this athletic empire. Newly independent republics lacked the resources to maintain elite training infrastructure, leading to decaying facilities, staff departures, and declining participation (O'Mahony 2006). Athlete representation fell from 481 in 1988 to 390 in 1996, with the overall share of total com-

[11] This local prominence is also evidenced in contemporary public engagement: as shown in Appendix Figure A1, cities with medal-winning athletes recorded significantly higher Baidu search volumes for "Olympics" during the 2024 Paris Games than cities without medalists, indicating that local communities follow the Games more closely when their own athletes win medals, underscoring the special significance these medals carry for their hometowns. Baidu is China's dominant search engine; its Search Index, functionally analogous to Google Trends, is widely used as a proxy for Chinese public attention.

petitors dropping from 5.7% to 3.8%, and women's participation declining even more sharply, from 7.4% to 4.7%. Because Olympic qualification demands strict performance benchmarks, these figures illustrate a sharp loss of competitive capacity.[12]

The Soviet collapse created a global vacuum, especially in state-intensive sports such as gymnastics, weightlifting, and track and field. As the centralized Soviet system disintegrated, the distribution of Olympic success shifted: China's total medal count doubled—from about 30 in Seoul 1988 to roughly 60 in Sydney 2000 and Athens 2004—while Soviet/Russian totals fell from about 130 to around 90. Over the same period, U.S. performance remained broadly stable at about 90–110 medals. Appendix Figure A2 illustrates this transition.

2.4. China's Educational System as a Multi-Tiered Tournament

The potential for female Olympic success to shape gender outcomes is especially salient in China's highly selective, exam-driven education system, where parental beliefs about children's competitive capacity strongly shape educational investment. The system functions as a multi-tiered tournament comprising four sequential stages: primary school (grades 1–6), junior high school (grades 7–9), senior high school (grades 10–12), and college. Advancement beyond junior high requires passing a high-stakes examination, the *Zhongkao* (High School Entrance Exam), with admission determined strictly by test scores. Students who advance to senior high school subsequently take the *Gaokao* (National College Entrance Exam) three years later—the sole pathway to college admission. These examinations act as high-stakes filters: only strong performers can progress to the next tier, making the system intensely competitive.

Parental decisions are concentrated at the junior- and senior-high transitions, where families weigh the costs of continued schooling against alternatives such as vocational training or early labor market entry. College admission, by contrast, turns almost entirely on *Gaokao* scores, leaving little room for parental discretion. Chinese parents often hold gender-asymmetric beliefs about competitive capacity that translate into investment patterns favoring sons over daughters (Qin, Stoddard, and Sojourner 2023; Kang, Lei, Song, and Zhang 2024).

3. Data

3.1. Data and Sample

We assemble several data sources spanning Olympic records, the population census, household and attitude surveys, media coverage, and supporting material for identifi-

[12]According to IOC and federation regulations, qualification depends on meeting event-specific performance standards or advancing through major tournaments such as World Championships.

cation:

Olympic Performance Data. We construct a comprehensive database of Chinese Summer Olympic participation from 1984 to 2016 using official records from the International Olympic Committee and supplementary sources including the Baidu Encyclopedia and Wikipedia. Our dataset contains 3,047 athlete-year-sport observations covering 2,264 unique athletes who won 782 medals. For each athlete, we collect demographic information including hometown, birth year, gender, physical characteristics, and competitive achievements such as medals and rankings.

Our main analysis focuses on six Summer Olympics from 1984 to 2004, excluding the 2008 Beijing Olympics due to home field advantage.[13] For robustness checks, we expand the sample to include the 2008 Beijing Olympics and additionally collect medal records and athlete information from China's participation in 12 Winter Olympics from 1980 to 2022.

Population Census. Our primary analytical data source is China's 2015 Population Mini-Census, a large-scale survey containing detailed information on approximately 1.3 million representative individuals. The census provides individual-level data on educational attainment, employment, occupation, and demographic characteristics, enabling us to construct measures of gender gaps across cohorts and hometown regions.

Social Attitudes and Behavior. We supplement census data with three nationally representative surveys, each offering a distinct but complementary aspects of the mechanisms we study. The Chinese General Social Survey (CGSS 2010) provides direct measures of general gender attitudes. The China Family Panel Studies (CFPS 2010) report parents' educational expectations for their children and children's self-reported attitudes, supporting our analysis of parental belief updating and children's self-perceptions. The Chinese Household Income Project (CHIP 1988, 1995, 1999) contains detailed household expenditure data that allow us to examine differential investment in children of different genders.

Media Coverage. To measure Olympic coverage intensity, we analyze 45,657 Olympic-related articles from the People's Daily (1946-2004), including 15,580 that specifically mention individual medalists. As China's primary state newspaper with nationwide circulation, the People's Daily provides a comprehensive measure of official media attention to Olympic achievements.

Other Complementary Datasets. We incorporate additional sources to strengthen our anal-

[13]We exclude the 2008 Beijing Olympics from the baseline analysis as an exceptional host-year episode rather than a typical realization of medal emergence. China won 100 medals and topped the gold-medal standings for the first (and so far only) time at the 2008 Beijing Olympics. Given the likely importance of host-country advantages in athlete selection, preparation, and performance, the emergence of medalists in 2008 is less plausibly exogenous than in earlier Games.

ysis. From China City Statistical Yearbooks, we extract prefecture-level socioeconomic indicators including per capita GDP and population. We collect county-level gazetteer books to identify Olympic-related events recorded in local historical records and utilize Baidu Search Index data to measure regional variation in public attention to Olympic competitions. To construct our instrumental variable, we compile medal records from China's 1983 National Games using official sports history archives and Soviet/Russia athlete participation data from International Olympic Committee registration records (1952-2004).

Sample Construction. We apply several restrictions to construct our analytical sample. We limit the sample to individuals born between 1960 and 1995, ensuring sufficient exposure to China's Olympic participation since 1984 for older cohorts while guaranteeing that younger cohorts completed their educational decisions by the 2015 census.[14] To ensure comparability across regions in our analysis, we focus on prefectures that produced at least one female Olympic medalist between 1984 and 2004, which yields a main sample of 258,857 individuals from 91 prefectures.

3.2. Variable Construction

Treatment Variables. Our primary treatment is *Medal Cohort*, a binary indicator equal to one if an individual was aged 18 or younger when at least one female athlete from their prefecture first won an Olympic medal (gold, silver, or bronze); we verify the robustness of our results to alternative age cutoffs of 12 and 15 in Section 5.1. We describe its construction and the underlying identifying variation in Section 4.1.

Educational Attainment Measures. We construct several complementary metrics from the population census data to measure educational achievement. First, we create a continuous measure of years of schooling by converting reported completed educational attainment into numerical values: 0 years for illiteracy, 6 years for completed primary school, 9 years for completed junior high school, 12 years for completed senior high school, 15 years for completed associate college, 16 years for completed four-year college, and 19 years for completed postgraduate education.

Second, we examine educational outcomes along two margins: transitions and dropouts. Transitions capture whether students advance from one educational level to the next, conditional on completing the previous stage. We construct binary indicators for: primary school enrollment or above, junior high enrollment (among primary completers), senior high enrollment (among junior high completers), college enrollment including associate and four-year college programs (among senior high completers). Dropouts measure exits before stage completion, conditional on enrollment. For each tier, dropout

[14] In robustness checks, we expand these sample restrictions and obtain consistent results.

equals one if an individual enrolled but did not complete that stage, and zero otherwise. We construct dropout indicators for four stages: primary school, junior high school, senior high school, and college.

Gender Attitudes. To explore potential mechanisms underlying Olympic effects, we construct measures of gender attitudes and household behavior using several representative household surveys. These measures allow us to examine whether exposure to female Olympic success shifts individual attitudes toward gender equality and influences household decision-making. We draw on CGSS survey questions that capture respondents' views on gender roles in society, the division of domestic responsibilities between spouses, and preferences over the gender of children. We draw from the CFPS parents' educational expectations for their children, together with children's self-reported expectations and aspirations. Additionally, we examine household expenditure patterns on education by child gender using CHIP data to assess whether exposure translates into differential investment behaviors.

Control Variables. We incorporate controls for individual characteristics such as ethnicity (Han or minority), household registration type (urban or rural hukou), and current migration status (whether hometown and current residence coincide). The inclusion of hukou status is particularly important given substantial educational disparities between urban and rural residents in China. We also incorporate prefecture-level controls to account for regional variation in economic development. From China City Statistical Yearbooks, we extract socioeconomic indicators including GDP per capita and population size to control for local economic conditions that may influence both Olympic performance and educational outcomes.

3.3. Summary statistics

This subsection summarizes the analytical sample and describes the treatment variation we exploit—the timing, sport composition, and geographic distribution of female Olympic medalists across Chinese prefectures between 1984 and 2004.

Appendix Table A1 presents summary statistics for our main sample of 258,857 individuals born between 1960 and 1995, drawn from China's 2015 Population Mini-Census. The sample is roughly gender-balanced (49.6% female), with 11.3% being cross-prefecture migrants who had moved away from their birth prefecture by 2015. Average educational attainment is 10.6 years of schooling, roughly corresponding to completion of junior high school and some senior high school. In terms of educational transitions, nearly all individuals (96%) entered primary school, 90% of primary school completers continued to junior high school, 49.7% of junior high school completers advanced to senior high school, and 52.3% of senior high school completers pursued higher education (including associate and four-year college programs). Dropout rates are low across all stages and decline with the level of education: 8.7% dropped out during primary school, followed by 2.8% during junior high school, 1.8% during senior high school, and 1% during college.

Appendix Figure A3 illustrates China's Olympic medal performance and female representation between 1984 and 2004. Total medals rose from 32 at the 1984 Los Angeles Olympics to 63 at the 2004 Athens Olympics, with particularly rapid growth after 1992 coinciding with the Soviet Union's dissolution. Notably, female athletes contributed a disproportionate share of China's Olympic success, with the female medal share rising to approximately 70% after 1992 and maintaining this high level across subsequent Olympic cycles.

Appendix Table A2 presents the top 10 Olympic sports ranked by the number of medals won by Chinese athletes between 1984 and 2004, along with the female share of medalists in each sport. Diving leads with the highest medal count, followed by gymnastics, shooting, and weightlifting, reflecting China's particular strengths in these disciplines. Examining female representation across these sports reveals substantial heterogeneity: women achieved complete dominance in certain disciplines–accounting for 100% of China's medals in swimming and judo, and over 80% in badminton and athletics–while representing considerably smaller proportions in others, such as weightlifting (24%) and gymnastics (30%).

Appendix Figure A4 illustrates the geographic and temporal distribution of China's first female Olympic medalists across prefecture-level administrative units between 1984 and 2004. The map reveals substantial spatial heterogeneity in the timing of female Olympic success, with the darkest shaded regions representing prefectures that produced their first female medalist in 1984 and progressively lighter shades indicating later years of first achievement. This staggered emergence of first female medalists across prefectures provides the key identifying variation for our empirical analysis, generating both cross-sectional and temporal variation in treatment assignment. Appendix Table A11 complements the map by listing all treated prefectures organized into six cohorts, each defined by the Olympic cycle in which the city first produced a female medalist.

4. Empirical Strategy

4.1. The Medal Cohort

We construct the treatment variable *Medal Cohort* to capture exposure to hometown female Olympic success during the educational choice window. This window denotes the years in which schooling decisions, and the parental investments that support them, are still being made and remain responsive to external influence. We define the variable

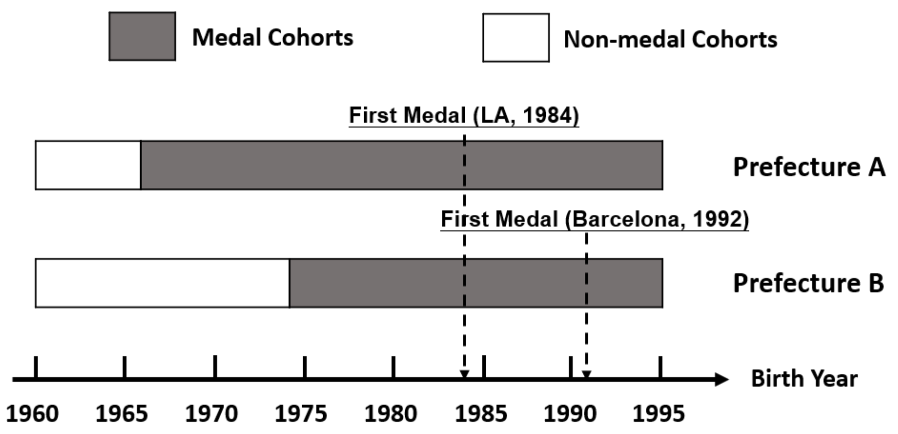


Figure 1. *Example of Empirical Strategy: Staggered Timing of First Female Olympic Medalists Across Prefectures. This figure illustrates our empirical strategy, which exploits two sources of variation to estimate the effects of a prefecture's first female Olympic success on gender-related educational outcomes. First, we use cross-prefecture variation in the timing of when different regions first produced female Olympic medalists (e.g., Prefecture A in the 1984 Los Angeles Olympics vs. Prefecture B in the 1992 Barcelona Olympics). Second, within each prefecture, we compare individuals from different birth cohorts who were at different stages of their educational decisions when the first female Olympic success occurred in their hometown.*

as a binary indicator equal to one for individuals who were aged 18 or younger when their prefecture first produced a female Olympic medalist (gold, silver, or bronze), and zero otherwise. The age-18 cutoff reflects the end of the period in which educational decisions are typically still being made; we test sensitivity to alternative cutoffs (ages 12 and 15) and to a continuous exposure measure in Section 5.1.

Figure 1 illustrates the resulting timing variation across prefectures. Consider two prefectures with different timing. Prefecture A first celebrated a female Olympic medalist in 1984, so individuals born between 1966 and 1995 in that prefecture were aged 18 or younger and are classified as *Medal Cohorts*; those born before 1966 are *Non-medal Cohorts*. Prefecture B reached the same milestone eight years later in 1992, so the corresponding Medal Cohorts are individuals born between 1974 and 1995. This staggered timing across prefectures, combined with within-prefecture variation across birth cohorts, provides the identifying variation we exploit in the specifications that follow.

Our empirical strategy focuses on medalists' hometown prefectures, where local pride, media coverage, and community recognition make their success most salient. Several patterns support this assumption. As documented in Section 2.2, medalists are widely celebrated in hometown county gazetteers, and during the Games, prefectures with hometown medalists show significantly higher Olympics-related search intensity than those without. Moreover, if the assumption holds true, we should expect the effect to be spatially concentrated: adjacent non-medal prefectures experience a smaller effect that decays with distance, a spillover pattern we examine in Section 5.1.

4.2. Fixed Effects Specification

We estimate the impact of female Olympic medalists on gender disparities in education using a fixed-effects specification that exploits the staggered timing variation defined in Section 4.1. Our benchmark sample focuses on the 91 prefectures that produced at least one female Olympic medalist between 1984 and 2004.

Our baseline specification is:

$$Y_{icp} = \beta_0 + \beta\,\text{Female}_i \times \text{Medal Cohort}_{cp} + \delta_{cp} + \gamma_{\text{Female}\times\text{Generation}} + X_i'\theta + \varepsilon_{icp}, \quad (1)$$

where Y_{icp} denotes the educational outcome for individual i born in calendar year c from prefecture p, and Medal Cohort$_{cp}$ is defined as in Section 4.1.

We include prefecture-by-birth-year fixed effects (δ_{cp}) to absorb prefecture-specific characteristics and birth-year factors shared by men and women alike. We also include interactions between the female indicator and generation-of-birth cohort dummies ($\gamma_{\text{Female}\times\text{Generation}}$) to flexibly capture gender differences across birth cohorts (i.e., the 1960s, 1970s, 1980s, and 1990s), capturing both secular changes in gender norms and the possibility that major institutional changes (such as nationwide education reforms) affected men and women differently across generations. We further include individual-level characteristics—ethnicity, hukou registration type, and migration status—to account for compositional differences across cohorts and prefectures. Standard errors are clustered at the prefecture level.

The coefficient β reflects the difference in gender education gaps between Medal and Non-medal Cohorts within the same prefecture and birth cohort, associated with the staggered timing of each prefecture's first female medal.

4.3. Shift-Share Instrument: Soviet Collapse and Medal Opportunities

Our fixed-effects specification absorbs time-invariant prefecture characteristics and birth-cohort-specific factors that could otherwise confound the gender-gap comparison. It does not, however, address the possibility that the timing of each prefecture's first female medalist—and hence the Medal Cohort variable defined in Section 4.1—is itself endogenous.

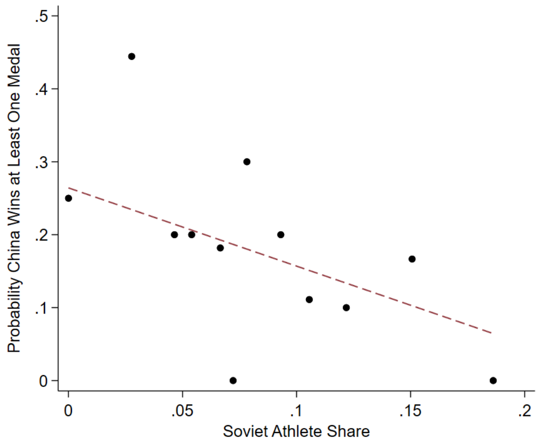
On the one hand, omitted variables–such as unobserved time-varying factors at regional level–might influence both the emergence of female Olympic medalists and local gender attitudes. For example, prefectures experiencing rapid economic growth or increasing cultural openness might simultaneously produce more female medalists and experience larger reductions in gender gaps. On the other hand, reverse causality may arise if regions with improving gender equality actively promote female participation in competitive sports, thereby increasing the likelihood of athletic success and confounding our interpretation.

To address these concerns, we implement an instrumental variables strategy that exploits plausibly exogenous variation from global geopolitical shocks that affected medal opportunities, thereby influencing the timing of the first female medalist across prefectures. Our instrument leverages the Soviet Union's historic Olympic dominance and the exogenous geopolitical events that weakened this dominance. As a leading competitor across numerous disciplines, Soviet participation–or absence–substantially reshaped medal opportunities for other countries. In China's case, geopolitical shocks such as the 1984 Soviet boycott of the Los Angeles Games and the Soviet Union's dissolution in 1991 altered medal prospects across sports in ways unrelated to local socioeconomic conditions or prevailing gender norms in Chinese prefectures. We exploit this externally induced variation in competitive opportunities as the basis for our identification strategy.

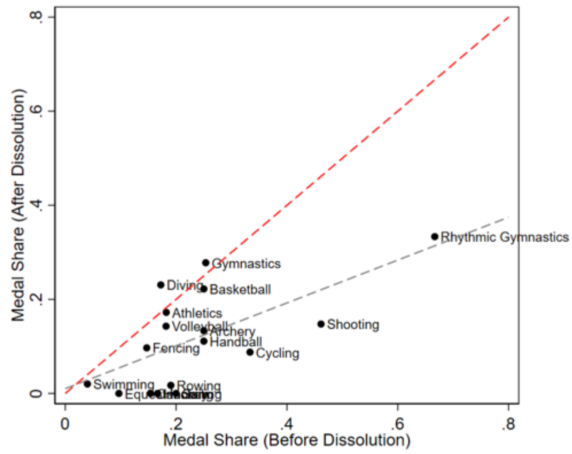
Our identification builds on a shift-share instrument that combines pre-determined athletic specializations of Chinese prefectures (the "shares") with exogenous changes in international competitiveness across sports disciplines driven by the Soviet Union's decline (the "shifts"). The intuition is straightforward: the collapse of the Soviet Union reshaped medal opportunities across sports in heterogeneous ways, and the extent to which each prefecture benefited depended on its historical sports composition. Following Borusyak, Hull, and Jaravel (2022), identification rests on the exogeneity of the *shifts*—sports-level changes in Soviet female participation—rather than the *shares*—prefecture-level specialization patterns.

The Soviet Union was historically among the most dominant competitors in the Olympic Games. However, its boycott of the 1984 Los Angeles Olympics and its dissolution in 1991 weakened this dominance and created a vacuum in the international competitive landscape. This vacuum shifted medal opportunities toward other countries, including China. Figure 2(a) provides evidence for this mechanism by examining the relationship between Soviet athlete participation and China's medal-winning probability across women's events. The binscatter plot reveals a strong negative correlation and demonstrates that events with higher Soviet participation rates were systematically associated with lower medal acquisition for Chinese athletes.

Crucially, the decline of Soviet competitiveness was uneven across disciplines. Sports in which the Soviet Union had historically invested heavily and consistently dominated experienced sharper declines in competition, whereas sports with limited Soviet pres-



(a) Soviet Share and China's Success



(b) Soviet Dominance Before vs After 1991

Figure 2. *Soviet Collapse and Chinese Olympic Opportunities. This Figure demonstrates the empirical basis for our instrumental variable strategy. Panel (a) shows a binscatter plot of China's medal-winning probability versus Soviet athlete participation rates (excluding China) across women's events from 1984-2004, controlling for sport fixed effects. The negative correlation confirms that Soviet dominance systematically reduced Chinese medal opportunities. Panel (b) compares Soviet medal shares in women's events before (1952-1988) and after (1996-2004) dissolution. The red dashed line marks the 45-degree line, while the gray dashed line shows the fitted relationship, revealing heterogeneous declines in Soviet performance across sports following political collapse.*

ence were much less affected. Figure 2(b) illustrates this pattern by comparing Soviet medal shares in women's events before (1952-1988, on the x-axis) and after dissolution (1996-2004, on the y-axis). The gray fitted line, with its slope significantly below unity, demonstrates that sports with initially higher Soviet dominance experienced disproportionately larger competitive declines following the collapse of Soviet Union. These cross-sport differences constitute the "shifts" in our shift-share construction.

The "shares" capture each prefecture's pre-determined athletic specialization, measured by performance in China's 1983 National Games–held before China's Olympic return and subsequent geopolitical shocks. These patterns reflect historical investments in facilities, coaching, and training systems, as well as established local talent pipelines.

The "shifts" are measured by the proportion of Soviet female athletes (excluding China) in each sport. Because Olympic participation requires meeting strict qualifying standards–such as world rankings or minimum scores–this proportion provides a credible proxy for Soviet competitive strength. Following the Soviet Union's boycott and subsequent dissolution, prefectures specialized in formerly Soviet-dominated sports suddenly faced greater medal opportunities, while those oriented toward unaffected sports saw little change. Interacting these predetermined shares with exogenous shifts generates plausibly exogenous variation in medal opportunities across prefectures.

Constructing IV. Having laid out the conceptual basis above, we now formalize the shift-share construction in five steps. First, to measure each prefecture's pre-determined athletic specialization, we collect information on female medalists from China's 1983 National Games–held before China's 1984 Olympic return. We record each medalist's sport discipline and hometown prefecture, then aggregate these data to construct hometown prefecture-sport specialization shares.

Specifically, for prefecture p in women's sport k, we define the specialization share:

$$\rho_{kp} = \frac{\text{Medals won by prefecture } p \text{ in women's sport } k \text{ at 1983 National Games}}{\text{Total medals in women's sport } k \text{ at 1983 National Games}}, \quad (2)$$

where higher values indicate stronger comparative advantage in sport k.[15]

Second, we compute the *shift* variable, s_{kt}, capturing Soviet competitive strength in sport k during Olympic year t (excluding Chinese athletes):

$$s_{kt} = \frac{\text{Soviet female athletes in sport } k \text{ at Olympics } t}{\text{Total female athletes in sport } k \text{ at Olympics } t \text{ (excluding China)}}. \quad (3)$$

Larger values indicate stronger Soviet presence and correspondingly fewer medal opportunities for others.

Third, we interact each prefecture's pre-determined specialization with these exogenous shocks:

$$\theta_{pt} = \sum_{k=1}^{K} \rho_{kp} \times s_{kt}. \quad (4)$$

This shift-share aggregation captures prefecture p's exposure to Soviet competition in year t. Lower values of θ_{pt} correspond to greater medal opportunities as Soviet dominance waned.

Fourth, we map θ_{pt} into a probability measure bounded between 0 and 1:

$$\pi_{pt} = 1 - e^{-(1-\theta_{pt})}. \quad (5)$$

We transform θ_{pt} into a probability bounded in $[0, 1]$ so that exposure can be aggregated across multiple Olympic years using the standard at-least-one-event formula in Equation (6), producing a cohort-level instrument on the same scale as the binary Medal Cohort variable defined in Section 4.1. The exponential form is motivated by interpreting medal opportunities as rare events in a Poisson process, where the rate parameter $(1 - \theta_{pt})$ reflects the intensity of competitive opportunities available to Chinese athletes

[15] For example, if women's shooting produced 10 medals at the 1983 National Games, with 5 medals won by athletes from Prefecture A, 3 from Prefecture B, and 2 from Prefecture C, then Prefecture A's strength in women's shooting would be 50%, Prefecture B's would be 30%, Prefecture C's would be 20%, and all other prefectures would have zero shares.

from prefecture p in Olympic year t, net of Soviet competitiveness. Higher values of θ_{pt} (stronger Soviet presence) therefore correspond to lower values of π_{pt}, reflecting that greater Soviet dominance reduces medal opportunities available to Chinese athletes from that prefecture.

Finally, we aggregate these opportunities into a cohort-level measure. For individuals born in prefecture p with birth year c, we define the *Predicted Medal Cohort* as cumulative exposure during ages 0-18:

$$\text{Predicted Medal Cohort}_{cp} = 1 - \prod_{j=1}^{J_c} (1 - \pi_{pt_j}), \tag{6}$$

where $t_j \in \{1984, 1988, 1992, 1996, 2000, 2004\}$ indexes Olympic years covered before age 18, and J_c is the number of such years. Intuitively, this represents the probability that an individual cohort experienced at least one medal opportunity from its home prefecture during the educational choice window.[16]

2SLS Regression. Using the predicted medal exposure, we employ a two-stage least squares estimation that proceeds as follows. In the first stage, we regress the observed medal cohort exposure on the predicted exposure generated by our shift-share instrument:

$$\begin{aligned} \text{Female}_i \times \text{Medal Cohort}_{cp} = \alpha_0 + \alpha_1\, & \text{Female}_i \times \text{Predicted Medal Cohort}_{cp} \\ & + \delta_{cp} + \gamma_{\text{Female}\times\text{Generation}} + X_i'\phi + \nu_{icp}, \end{aligned} \tag{7}$$

where Predicted Medal Cohort$_{pc}$ is constructed following Equation (6) using pre-determined prefecture athletic specialization interacted with time-varying Soviet participation rates. The coefficient α_1 captures the strength of the instrument, with larger values indicating that exogenous shifts in medal opportunities strongly predict actual medalist emergence.

In the second stage, we estimate the causal effect of female medalist exposure on educational outcomes using the predicted values from the first stage:

$$Y_{icp} = \beta_0 + \beta\, \text{Female}_i \times \widehat{\text{Medal}}\ \text{Cohort}_{cp} + \delta_{cp} + \gamma_{\text{Female}\times\text{Generation}} + X_i'\theta + \varepsilon_{icp}, \tag{8}$$

where Female$_i \times \widehat{\text{Medal}}$ Cohort$_{cp}$ denotes the fitted values from Equation (7). The coefficient β represents the local average treatment effect of exposure to female Olympic medalists on gender gaps in educational attainment, identified through exogenous variation in medal opportunities created by geopolitical shocks to international compe-

[16] For example, a cohort born in 1979 would be exposed to the 1984, 1988, 1992, and 1996 Olympics before age 18. If the prefecture's predicted medal probabilities in these years were 0.1, 0.1, 0.2, and 0.2, respectively, the predicted medal exposure would be $1 - (1 - 0.1)(1 - 0.1)(1 - 0.2)(1 - 0.2) = 0.4816$.

tition.

Exclusion Restriction. As noted earlier, our identification rests on shift exogeneity. The exclusion restriction therefore requires that Soviet participation influence gender outcomes in Chinese prefectures only through its effect on medal opportunities, with no independent impact through channels such as local gender attitudes.

This assumption is plausible because Soviet participation was determined by domestic sports planning and broader geopolitical factors unrelated to the evolution of gender stereotypes in China. We report a set of validation exercises supporting this identifying assumption in Section 5.4, with detailed results in Appendix C.

5. Empirical Results

5.1. Fixed Effects Estimation

Table 1 presents our baseline fixed-effects estimates, with years of schooling as the outcome. The key coefficient of interest is the interaction term Female $\times$ Medal Cohort, which captures the differential impact of female medalist exposure on women's educational outcomes relative to men's.

The results reveal a consistent and statistically significant positive effect across all specifications. The key estimated coefficient ranges from 0.198 to 0.2, indicating that females exposed to female Olympic medalists from their home prefecture during the educational choice window complete approximately 0.2 additional years of schooling compared to their male peers. This represents roughly a 2% increase relative to the sample mean of 10.65 years of schooling and roughly one-fifth of the baseline gender gap of 0.99 years (calculated as the average difference in years of schooling between males and females among pre-Olympic cohorts), suggesting an economically meaningful impact of female medalist exposure on educational attainment.

These findings are robust across different model specifications. Column (1) includes basic fixed effects and generation-of-birth fixed effects, while columns (2) and (3) add increasingly stringent controls, including the interactions between home prefecture and birth year, as well as a comprehensive set of individual characteristics.

As a placebo test, we construct an analogous *Male Medal Cohort* variable capturing exposure to hometown male Olympic medalists during the educational choice window. If the gains reflected general Olympic success rather than female-specific achievement, this exposure should produce comparable effects on women's outcomes. Column (4) of Table 1 reports both Female $\times$ Medal Cohort and Female $\times$ Male Medal Cohort: the female-medalist effect remains robust while the male-medalist coefficient is small and statistically insignificant. The pattern itself is suggestive—only the kind of success that

Table 1. *Effect of Female Olympic Medal on Gender Disparities in Years of Schooling*

	Dependent Variable: Years of Schooling			
	Prefectures with Female Medalist			Also Male Medalist
	(1)	(2)	(3)	(4)
Female × Medal Cohort	0.198***	0.199***	0.200***	0.225**
	(0.0641)	(0.0637)	(0.0635)	(0.109)
Medal Cohort	-0.124*			
	(0.0713)			
Female × Male Medal Cohort				-0.0285
				(0.0826)
Controls			Y	Y
HomePref FE	Y			
BirthYear FE	Y			
HomePref×BirthYear		Y	Y	Y
Female×Generation	Y	Y	Y	Y
Mean	10.65	10.65	10.65	11.50
N	258,857	258,857	258,857	113,441
R^2	0.248	0.268	0.270	0.258

Notes: This table presents the baseline estimates of the effect of exposure to hometown female Olympic medalists on gender disparities in years of schooling. Column (1) includes prefecture fixed effects, birth year fixed effects, and the Female × Generation interaction term. Column (2) adds individual-level controls including an indicator for Han ethnicity, an indicator for urban hukou status, and an indicator for migration status. Column (3) is our preferred specification, which replaces the separate prefecture and birth year fixed effects with prefecture-by-birth year fixed effects while retaining all individual-level controls from column (2). Column (4) restricts the sample to prefectures that produced both a female medalist and a male medalist, and includes an additional interaction term Female × Male Medal Cohort to directly compare the gender-specific effects of female versus male medalist exposure. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

directly contests a stereotype about women in particular appears to produce effects.

We also show that our findings are robust to alternative treatment definitions. Appendix Table A3 tests sensitivity to the age cutoff for defining *Medal Cohort* (using thresholds of 12 and 15 years instead of 18) and replaces the binary indicator with a continuous measure capturing the number of years each individual was exposed to female Olympic medalists before age 18. All alternative measures yield consistent results.

We address several additional concerns about sample composition, treatment definition, and time-varying confounders in Appendix Table A4.

Sample composition. (1) The 1984 Los Angeles Olympics was exceptional due to the Soviet-led boycott; we exclude prefectures that produced their first female medalist in 1984 to ensure our results are not driven by this unique event. (2) We expand the sample to include all prefectures (both medal and non-medal) and all birth cohorts, addressing potential sample selection concerns. (3) We exclude individuals whose residence differs from their birthplace to ensure accurate measurement of medalist exposure.

Treatment definition. (4) We extend the medalist sample to include female medalists from the 2008 Beijing Summer Olympics, extending treated status to prefectures whose first female Olympic medalist competed in or before 2008. (5) We further extend the definition of medalist exposure to include Winter Olympics medalists, reassigning treatment based on each prefecture's first female medalist across both Summer and Winter Games.

Additional controls. (6) We control for compulsory education policy reforms to account for concurrent educational expansions that may confound our estimates. (7) We include gender-by-prefecture fixed effects to absorb region-specific gender differences. (8) To address time-varying factors that differentially affect male and female education, we interact the female indicator with regional GDP per capita and the number of primary schools, imputing prefecture-year missing values using the corresponding provincial average. Across all specifications, the estimated effects remain statistically significant and similar in magnitude.

Geographic Spillovers. We next examine whether the hometown female-medalist effect diffuses to nearby non-medal areas. Focusing on non-medal prefectures, we construct two samples that differ in their proximity to medal prefectures: *exposed neighbors*, the non-medal prefectures adjacent to medal prefectures, and *placebo neighbors*, the non-medal prefectures adjacent to these *exposed neighbors* but not bordering any medal prefecture. Intuitively, the exposed neighbors form the first ring directly surrounding the medal prefectures, while the placebo neighbors lie one ring farther out. For each non-medal prefecture, we define treatment exposure using the first female medalist from its *nearest* medal prefecture. Appendix Table A5 reports the results. For exposed neighbors (column (1)), the coefficient is positive and economically meaningful—roughly two-thirds of our baseline estimate—though it falls short of statistical significance. For the more distant placebo neighbors (column (2)), the effect is markedly smaller and close to zero. This gradient—sizable in prefectures adjacent to treated ones and fading with distance—is consistent with a localized spillover and reinforces our focus on medalists' own hometowns.

Heterogeneity by Sport Type. If the educational gains arise because female success challenges stereotypes about women's *competitive* capacity rather than their athletic talent per se, the effect should be strongest in disciplines that are more confrontational or more reliant on strength. We test this by splitting sports along two dimensions. First, we contrast head-to-head sports (e.g., basketball), where competitive prowess is on direct display, with performance-based sports (e.g., diving), judged against fixed standards. Second, we contrast strength-oriented disciplines (e.g., weightlifting), which rebut the belief that women are physically weaker, with skill-oriented disciplines (e.g., rhythmic gymnastics), more readily reconciled with stereotypes of female grace. Appendix Table A6 confirms both predictions: head-to-head and strength-oriented success yield substantially larger effects than their performance-based and skill-oriented counterparts. This reinforces our interpretation that it is achievement demonstrating women's capacity to compete and prevail, not female athletic success in general, that most effectively narrows the educational gender gap.

5.2. More Transitions or Fewer Dropouts?

Having established the main effect, we ask whether it comes from more girls advancing to higher tiers (*transitions*) or fewer girls dropping out once enrolled (*dropouts*). We examine each channel in turn.

We first focus on transitions. We estimate Equation (1) separately on four samples—the full sample, primary-school completers, junior-high completers, and senior-high completers—with the outcome in each case an indicator for enrollment at the next tier (defined in Section 3.2).

Table 2 presents the results. Female medalist exposure raises girls' enrollment at every transition prior to college (relative to boys), with the largest effects at the high-stakes secondary transitions, consistent with evidence that gender differences in competitiveness become especially consequential at pivotal choice points (Buser, Niederle, and Oosterbeek 2014). The estimated effect is 0.94 percentage points at primary school (column 1; 1% of mean), 2.64 percentage points at junior high conditional on primary completion (column 2; 2.9%), and 1.19 percentage points at senior high conditional on junior-high completion (column 3; 2.4%). Beyond high school, the effect attenuates: the coefficient for college entry conditional on senior-high completion remains positive but insignificant (column 4).

We next examine dropouts. For each educational stage, we restrict the sample to students who enrolled and estimate whether they completed that stage; the dependent variable is a binary indicator equal to one if the individual dropped out before completion. Table 3 presents the results. Conditional on enrollment, we find no significant impact on dropout rates at any educational stage.

Taken together, the gains in women's educational attainment come from advancement at the secondary transitions, not from reduced dropout. The contrast—effects where parental investment decisions matter most (junior- and senior-high tracking, where families weigh continued schooling against vocational alternatives) and absence where they do not (college admission, determined by *Gaokao* scores)—points to parental decision-making as the operative channel, a mechanism we develop in Section 6.2.

Table 2. *Effect of Female Olympic Medal on Gender Disparities in Educational Transitions*

	Dependent Variable: Educational Transitions			
	Illiteracy→Primary	Primary→Junior	Junior→High	High→College
	(1)	(2)	(3)	(4)
Female × Medal Cohort	0.00939**	0.0264***	0.0119*	0.0112
	(0.00377)	(0.00743)	(0.00701)	(0.00984)
Controls	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y
Mean	0.960	0.900	0.497	0.523
N	258,857	248,512	223,585	111,035
R^2	0.049	0.132	0.178	0.157

Notes: This table examines the effect of exposure to hometown female Olympic medalists on gender disparities in educational transitions at different stages. Column (1) uses the full sample and examines whether individuals enrolled in primary school or above. Column (2) restricts to individuals who completed primary school and examines enrollment in junior high school or above. Column (3) restricts to individuals who completed junior high school and examines enrollment in senior high school or above. Column (4) restricts to individuals who completed senior high school and examines enrollment in associate college, college or above. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

5.3. Shift-Share IV Estimates

We now report the IV estimates. Recall from Section 4.3 that our shift-share instrument exploits the post-1991 Soviet collapse, which generated exogenous variation in medal opportunities across Chinese prefectures depending on their pre-determined sports specializations from the 1983 National Games.

Table 4 presents the IV estimates. To assess instrument strength, we regress observed medal exposure on the shift-share-predicted measure in the first stage. Column (1) reports a coefficient of 0.59 with an F-statistic of 40.3, well above conventional weak-instrument thresholds. The second-stage estimates in column (2) show that medal exposure reduces the educational gender gap by approximately 0.56 years, an effect that is statistically significant and economically meaningful.

As in our Section 5.1 placebo, we construct an analogous shift-share IV for male medal cohorts, using male-athlete shares from the 1983 National Games and the same Soviet-shift series. If the educational effects reflect general Olympic visibility rather than female-specific success, this exposure should produce comparable effects on women's outcomes. Columns (3) and (4) of Table 4 present the results. The first-stage relationship for male medals (column 3) remains positive and statistically significant, as expected: the same geopolitical shocks and athletic specialization patterns that created

Table 3. Effect of Female Olympic Medal on Gender Disparities in Educational Dropout

	Dependent Variable: Dropout (1=Yes)			
	Primary Dropout	Junior Dropout	High Dropout	College Dropout
	(1)	(2)	(3)	(4)
Female × Medal Cohort	0.00363	0.00169	-0.00192	0.00221
	(0.00994)	(0.00172)	(0.00303)	(0.00293)
Controls	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y
Mean	0.087	0.028	0.018	0.010
N	26,912	131,473	66,707	58,536
R^2	0.111	0.035	0.061	0.092

Notes: This table examines the effect of exposure to hometown female Olympic medalists on gender disparities in educational dropout at different stages. The dependent variable is a binary indicator equal to one if an individual entered but did not complete a specific educational stage (i.e., dropped out). Column (1) examines dropout among those who entered primary school. Column (2) examines dropout among those who entered junior high school. Column (3) examines dropout among those who entered senior high school. Column (4) examines dropout among those who entered an associate college or above. Each column conditions on entry into the respective educational level and measures whether the individual failed to complete it. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses are clustered at the prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

opportunities for female athletes should similarly affect male athletes.

The second-stage results, however, reveal a crucial asymmetry (column 4): male medal cohort exposure shows no significant effect on the educational gender gap, mirroring the FE placebo in column (4) of Table 1. This helps mitigate concerns that unobserved prefecture-specific trends correlate with both Olympic success and gender outcomes.

5.4. IV Robustness Checks

Exogeneity of the Shifts. The key identification concern is that the Soviet retreat from women's sports may have been systematically aligned with China's emerging strengths rather than distributed quasi-randomly across disciplines. If Soviet athletes disproportionately reduced participation in sports where Chinese competitors were gaining ground, our estimated shifts would be endogenous to China's athletic specialization.

To assess this, we examine whether Soviet decline correlates with China's participation in the two Olympic Games held immediately before the dissolution. Specifically, we construct changes in the Soviet share of female athletes between 1988 and 1996 across sports and regress these on China's athlete shares in 1984 and 1988, respectively. Appendix Figure A6(a) presents results using the 1984 Los Angeles Games, while Appendix Figure A6(b) uses the 1988 Seoul Games.

Table 4. Bartik IV Regression: Female vs Male Medal Effects

	Female Medal		Male Medal	
	1^{st} Stage	2^{nd} Stage	1^{st} Stage	2^{nd} Stage
Dependent Variable	Female×Medal Cohort	Years of Schooling	Female×Medal Cohort	Years of Schooling
	(1)	(2)	(3)	(4)
Female × Medal Cohort		0.558***		
		(0.173)		
Female × Predicted Medal Cohort	0.591***			
	(0.0932)			
Female × Male Medal Cohort				0.236
				(0.431)
Female × Predicted Male Medal Cohort			0.411**	
			(0.153)	
Controls	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y
F-statistics		40.299		7.273
N	258,857	258,857	113,441	113,441
R^2	0.892	0.003	0.879	0.002

Notes: This table presents instrumental variable estimates using a Bartik-style instrument that interacts prefecture-level pre-determined athletic specialization with time-varying Soviet female athlete participation rates across Olympic sports. Columns (1)-(2) examine female medalist effects: column (1) shows the first-stage relationship between the instrument (Female × Predicted Medal Cohort) and actual medal exposure (Female × Medal Cohort), while column (2) presents the second-stage effect on years of schooling. Columns (3)-(4) present analogous specifications for male medalist exposure as a placebo test. The F-statistics reported are Kleibergen-Paap Wald F-statistics. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Across both specifications, we find near-zero correlations: the Soviet withdrawal was uncorrelated with China's pre-existing strengths, supporting the assumption that our shifts capture plausibly exogenous variation in competitive opportunities.

Balance Test. Even if shifts are exogenous at the sport level, the instrument could still violate the exclusion restriction if it systematically predicts pre-determined regional characteristics that shape gender stereotypes or educational outcomes. To test this, Appendix Table A12 reports comprehensive prefecture-level balance checks. We assess whether predicted medal opportunities for the 1992 cohort correlate with key pre-dissolution characteristics: the sex ratio (female share of population), GDP per capita, primary and secondary industry shares, and the density of primary schools, gymnasiums, and hospitals per 1,000 residents.

Across all dimensions, we find no significant correlations. This indicates that our instrument captures variation orthogonal to regional characteristics that might otherwise explain changes in educational outcomes.

Controlling for Dynamics of Regional Characteristics. A further concern is that regional characteristics influencing education may evolve over time in ways that spuriously correlate with medal opportunities, even if baseline levels are balanced. To address this, Appendix Table A13 reports results controlling for interactions between pre-1991 prefecture characteristics and birth cohort fixed effects. The main coefficients remain stable under these specifications, suggesting that time-varying regional heterogeneity does not drive our findings.

Pre-trend Test. A key concern is that regional characteristics shaping medal opportunities may also correlate with unobserved long-term factors that jointly influence educational outcomes, creating spurious effects. If this were the case, our predicted medal opportunities for the 1990s cohorts should also correlate with outcomes of earlier cohorts that were unaffected by the Soviet decline in women's sports.

To test this, Appendix Table A14 reports a placebo analysis using cohorts born in the 1960s-1980s. For each individual in these earlier cohorts (e.g., 1965), we assign the predicted medal opportunities calculated for the corresponding prefecture and year of a later 1990s cohort (e.g., 1995). The absence of significant effects in these regressions, together with weak first-stage relationships, suggests that our instrument captures the impact of the Soviet decline in international competitiveness rather than reflecting pre-existing educational trends.

Many-Shifts Requirement. As Borusyak, Hull, and Jaravel (2025) emphasize, the exogenous-shifts approach requires sufficient shifts to ensure the law of large numbers applies: if the shift number is too small, the shifts may by chance be correlated with unobservables even if they are truly random. Appendix Table A15 addresses this requirement by reconstructing our instrument using 214 detailed Olympic events rather than 29 comprehensive sport categories, substantially increasing the effective number of shifts. The results remain consistent under this finer categorization, indicating that our identification is unlikely to suffer from the few-shifts problem.

Alternative Shifts. In the benchmark analysis, we use Russian participation as the proxy for post-Soviet competitive strength. As an alternative, we construct a proxy based on the combined participation of Russia, Ukraine, and Belarus—the only other former republics with substantial Olympic presence and medal success during our study period. Appendix Table A16 shows that the coefficients remain highly similar across the two specifications, indicating that our results are not sensitive to the precise definition of the shift variable. The robustness is explained by the strong correlation in female athlete participation shares across sports when comparing Russia alone with the Russia–Ukraine–Belarus aggregate (Appendix Figure A7).

Recentering the Instrument. As emphasized by Borusyak and Hull (2023), the predetermined shares may induce omitted-variable bias even when the shifts are exogenous, a concern further sharpened by the nonlinearity of our instrument relative to the standard linear shift-share specification. We implement their recentering procedure, which subtracts from the instrument the expected value of the instrument conditional on the prefecture's share vector, isolating variation driven by the shifts alone.[17]

Appendix Table A17 reports the results. The first stage (column (1)) confirms that the recentered instrument remains a strong predictor of actual medal exposure, with an F-statistic of 21.7. The second-stage estimate (column (2)) yields a coefficient of 0.384, statistically significant at the 5% level and comparable in magnitude to our baseline IV estimate. We further conduct randomization inference based on the same 2,000 permutations to assess significance without relying on asymptotic approximations; the resulting p-value of 0.03 confirms that the estimate is unlikely to arise by chance (Appendix Figure A5). These results indicate that our findings are not driven by the mechanical component of the instrument that is predetermined by the share structure, reinforcing the causal interpretation.

6. Mechanism Analysis

Having shown that exposure to female Olympic medalists reduces educational gender gaps, we now turn to the underlying mechanisms. Among the many channels through which the effect may manifest, we provide evidence for two complementary ones: (i) media coverage that frames female athletic success as evidence of competitive capacity, and (ii) shifts in parental beliefs and household investment in daughters. We examine these channels using content analysis of *People's Daily*, the China Family Panel Studies (CFPS), the China Household Income Project (CHIP), and the Chinese General Social Survey (CGSS).

6.1. Media Coverage

We begin by examining the media channel through which female Olympic success may influence gender beliefs. In China, sporting events are subject to a distinctive state-media propaganda culture: athletic achievements on the international stage are routinely framed as evidence of *national pride* (an artifact of nationalism), while female athletic success is additionally framed as evidence of *competitive equality* (an artifact of

[17]Formally, the recentered instrument is $\tilde{z}_{cp} = z_{cp} - \mu_{cp}$, where $\mu_{cp} = \mathbb{E}[z_{cp} \mid \boldsymbol{\rho}_p]$. Because the instrument is a non-linear function of the shocks, μ_{cp} has no closed-form expression and must be approximated by simulation. Treating Soviet female participation rates as exchangeable across sports within each Olympic year, we generate $S = 2{,}000$ counterfactual instruments by permuting $\{s_{kt}\}_k$ within year while holding prefecture shares fixed, and set $\hat{\mu}_{cp}$ equal to the average across draws.

communism's foundational commitment to women's emancipation).

These dual framings appear vividly in *People's Daily* coverage. Following Xu Haifeng's gold medal in shooting at the 1984 Los Angeles Olympics—China's first-ever Olympic gold—the paper celebrated his victory as bringing "glory to our motherland" and marking "China's breakthrough on the world sporting stage."[18] Following Deng Yaping's domination of Olympic women's table tennis (gold medals at both the 1992 and 1996 Games), the same paper invoked "women have held up half the sky" to portray China's female athletes as evidence of women's equal capability with men.[19]

Our hypothesis is that this gender-equality framing—a by-product of state propaganda rather than its primary intent—challenged traditional stereotypes about women's competitive capacity, with downstream consequences for educational outcomes. Testing this hypothesis requires focusing on state media: in an era before widespread internet access and social media (1980s–2000s), state-controlled outlets were the dominant—and often sole—source of public information, giving official narratives exceptional influence over how citizens interpreted current events.

We analyze textual data from *People's Daily*, the principal propaganda organ of the Chinese Communist Party (CCP). Three features make it well suited for our purpose. First, during our study period, China's media system remained dominated by Party-state outlets; although commercialization gradually emerged during the reform era, it had not displaced official newspapers as the primary vehicles for political and social narratives. Second, sitting atop the official media hierarchy, *People's Daily* set the tone for news framing nationwide: provincial and local state media routinely republished, cited, or echoed its reports and editorials, allowing its narratives to reach local audiences. Third, *People's Daily* is a centrally edited national newspaper, with editorial decisions made in Beijing rather than locally; this institutional separation alleviates the concern that local gender attitudes shaped the framing of female athletic achievements rather than the reverse. Together, these features make *People's Daily* a particularly useful source for measuring the official framing of Olympic success and the narratives that plausibly shaped public beliefs across localities.

We test two key questions. First, did media coverage of male and female medalists differ in frequency or narrative framing? Second, did the impact of female medalists stem primarily from *thematic framing*–particularly when reports emphasized women's

[18] *People's Daily*, July 30, 1984: *"Xu Haifeng's historic victory brings glory to our motherland, marking China's breakthrough on the world sporting stage..."* Similarly, after Li Ning's three gold medals in gymnastics at the same Games, coverage stressed that "Li Ning's exceptional performance demonstrates our nation's strength and elevates China's international standing in the global arena" (*People's Daily*, August 6, 1984).

[19] *People's Daily*, August 2, 1996: *"Heroic women, admired by the world, have held up half the sky on China's gold medal tally..."* Similarly, after Zhang Shan won the mixed-gender skeet-shooting event at the 1992 Barcelona Olympics by defeating all male and female competitors, the paper celebrated her victory with the phrase "women are not inferior to men" (*People's Daily*, August 14, 1992).

competitive capacity and success in high-stakes contests–or simply from the *volume* of coverage they received?

Data and Measurement. Our analysis covers Chinese Olympic medalists from 1984–2004, yielding 416 medal records from 355 unique medalists, including 254 women and 101 men. For each medalist, we collect all *People's Daily* articles published after they won their medal. Among these, 15,580 articles mention Olympic athletes by name, forming our main dataset for analysis.

To systematically identify thematic framing in *People's Daily* coverage, we employ a supervised machine-learning approach. We segment 15,580 athlete-related articles into 91,507 sentences and fine-tune a pre-trained RoBERTa language model to classify each sentence into one of two themes. For competitive equality, we train on 500 sentences containing competitive-equality discourse (e.g., "women hold up half the sky," "women are not inferior to men") and 5,000 unrelated controls; for national pride, a placebo theme, we train on 500 sentences expressing national pride (e.g., "glory for the motherland," "national honor") and 5,000 unrelated controls. An article is labeled as belonging to a theme if it contains at least one classified sentence of that theme. Both classifiers achieve over 95 percent validation accuracy: the competitive-equality classifier identifies 1,963 sentences across 1,070 articles, and the national-pride classifier identifies 2,348 sentences across 1,304 articles. Further technical details appear in Appendix D.

These classifications enable the construction of athlete-year–level measures of thematic emphasis across time and individuals. We aggregate article-level observations to the athlete level and generate three key indicators for each medalist: (i) coverage intensity–the total number of *People's Daily* articles featuring the athlete following an Olympic medal; (ii) competitive-equality content share–the proportion of an athlete's coverage discussing competitive-equality themes; and (iii) national-pride content share–the proportion of coverage emphasizing national-pride narratives. These variables allow us to trace how both the scale and framing of media attention vary systematically across athletes and over time.

Gender Differences in Media Coverage. We first examine whether there are significant differences between female and male medalists in coverage intensity and thematic content.

$$y_{ikt} = \alpha + \beta \cdot \text{Female Medalist}_i + \gamma_k + \delta_t + \epsilon_{ikt}. \tag{9}$$

where y_{ikt} represents three measures for medalist i in sport k medal year t: coverage intensity, competitive equality share, and national pride share. Female Medalist$_i$ is an indicator for female athletes, while γ_k and δ_t capture sport and medal year fixed effects, respectively.

Table 5. Gender Differences in Olympic Medalists' Media Coverage

	Dependent Variable:		
	Number of Reports (log)	Competitive Equality Share	National Pride Share
	(1)	(2)	(3)
Female Medalist	-0.0118	0.0723***	0.0132
	(0.106)	(0.0148)	(0.0180)
Sport FE	Y	Y	Y
Year FE	Y	Y	Y
Mean	1.85	0.11	0.14
N	416	416	416
R^2	0.704	0.544	0.588

Notes: This table examines gender differences in Olympic medalists' media coverage characteristics using People's Daily article data. The dependent variables are the total number of articles mentioning each athlete in logarithm (column (1)), the share of coverage containing gender competitive equality themes identified through supervised machine learning (column (2)), and the share containing national pride themes (column (3)). All specifications include sport fixed effects and Olympic year fixed effects. Standard errors in parentheses; * p<0.1, ** p<0.05, *** p<0.01.

Table 5 presents the results. We find that female and male medalists receive comparable coverage volume, with column (1) showing no statistically significant difference in article counts. However, the content of coverage differs systematically by gender. Female medalists' coverage is significantly more likely to emphasize competitive equality themes (column (2)), with the share of such content being approximately 7.2 percentage points higher (equivalent to 66% of the sample mean). In contrast, national pride content shows no significant gender difference (column (3)), suggesting that both male and female medalists are equally likely to be framed in patriotic terms.

Heterogeneity by Coverage Characteristics. We next examine whether the impact of female medalists varies with media coverage characteristics–specifically, whether effects are driven by reporting intensity or by thematic framing. If media serve as the transmission channel for changing gender stereotypes, larger effects should arise from athletes whose coverage stresses female's competitive capacity narratives rather than from those who simply receive more attention. In contrast, narratives centered on national pride should not exhibit similar heterogeneity, as they do not directly challenge gender stereotypes or promote women's empowerment.

We assess these predictions through subsample analysis. Athletes are divided into high- and low-coverage groups based on whether their *People's Daily* coverage in the medal-winning year exceeds the sample median for each attribute—reporting intensity, competitive-equality content share, and national-pride content share. We then estimate the baseline specification (Equation (1)) separately for each subgroup. Table 6 summarizes the results along these three dimensions. Coverage intensity does not substantially influence the estimated effects: both high- and low-coverage female medalists generate similar positive impacts. By contrast, thematic framing matters considerably. Female medalists whose media portrayals place greater emphasis on competitive-equality themes exhibit markedly larger effects than those with weaker competitive-equality framing. National-pride framing, in turn, displays little heterogeneity: estimated effects are broadly similar across high- and low-content groups.[20]

Together, these findings indicate that media narratives, rather than exposure volume alone, may be responsible for the observed local gender responses. When state coverage explicitly links athletic achievement to women's broader capabilities—directly challenging prevailing stereotypes about female potential—the narrative may seep in, change minds, and become socially transformative. We turn next to the household side, where this framing plausibly translates into shifts in parental beliefs and educational investment.

6.2. Shifts in Parental Beliefs and Educational Investment in Daughters

We now explore the second mechanism: whether the observed educational effects operate through shifts in parental beliefs and household investment in daughters. The underlying hypothesis is that salient examples of hometown female athletes succeeding in high-stakes competitions reshape parents' stereotypes about their daughters' competitive capacity, leading to higher educational expectations for girls and stronger investment in their schooling. We test this hypothesis along three complementary dimensions using nationally representative survey data. We first examine whether hometown female-medalist exposure shifts parents' educational expectations for daughters relative to sons. We then examine whether the same exposure narrows the daughter–son gap in education-related household spending. Finally, we test whether the effects might instead operate through girls' own self-beliefs by examining children's self-reported attitudes and aspirations.

We begin by examining whether the emergence of female Olympic medalists was associated with shifts in parental education expectations using data from the 2010 China Family Panel Studies (CFPS), a nationally representative survey covering 128 prefecture-level cities. The CFPS surveys parents of children aged 0–15 and collects detailed information on each child's demographics, education, living conditions, health, and parental beliefs about childrearing. Our analysis includes all surveyed cities in the

[20]In addition to the subgroup-specific regressions, we estimate pooled specifications augmented with triple interaction terms, *Female* $\times$ *Medal Cohort* $\times$ *High Type*, where *High Type* is an indicator equal to one if an athlete's coverage exceeds the sample median for the relevant dimension. Only the interaction along the competitive-equality dimension attains statistical significance; the corresponding interactions for reporting intensity and national-pride content do not.

Table 6. People's Daily Coverage and Female Medalist Exposure

Group by	Dependent Variable: Years of Schooling					
	Number of Reports		Gender Equality		National Pride	
	High	Low	High	Low	High	Low
	(1)	(2)	(3)	(4)	(5)	(6)
Female×Medal Cohort	0.210**	0.204**	0.261***	0.0739	0.210**	0.186**
	(0.0823)	(0.0986)	(0.0675)	(0.129)	(0.0956)	(0.0834)
Controls	Y	Y	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y	Y	Y
Mean	10.56	10.82	10.45	11.15	10.49	10.91
N	172,135	86,722	186,864	71,993	163,100	95,757
R^2	0.253	0.297	0.266	0.261	0.265	0.273

Notes: This table examines heterogeneity in the effect of exposure to hometown female Olympic medalists by media coverage characteristics. The dependent variable is years of schooling. Columns (1)-(2) split the sample by coverage intensity: column (1) restricts to individuals exposed to high-coverage medalists (above median number of People's Daily articles in the medal-winning year), and column (2) to low-coverage medalists (below median). Columns (3)-(4) repeat this analysis splitting by the share of gender equality content in coverage during the medal-winning year, while columns (5)-(6) split by the share of national pride content in coverage during the medal-winning year. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

full sample.

The CFPS elicits parental educational aspirations by asking parents to indicate the highest level of education they hope their child will attain. Response categories range from "illiterate/semi-literate" to "doctoral degree," with intermediate options including primary school, junior high school, senior high school, associate college, bachelor's degree, and master's degree. We convert these responses into years of schooling and construct binary indicators corresponding to key educational thresholds. To test whether parental aspirations differ by child gender, we compare expectations for daughters and sons.

We construct Medal Exposure$_p$, a binary variable equal to 1 if prefecture p, where the household is located, had produced at least one female Olympic medalist before the survey year, and 0 otherwise. This variable captures whether parents were exposed to a hometown female Olympic medalist. We estimate the following specification:

$$Expectation_{icp} = \beta_1 \text{Daughter}_i + \beta_2 \text{Daughter}_i \times \text{Medal Exposure}_p + X_i'\theta + \delta_p + \lambda_c + \epsilon_{icp}, \quad (10)$$

where $Expectation_{icp}$ denotes the educational expectation held by a parent (father or mother) born in year c and residing in prefecture p for child i. Daughter$_i$ is an indicator for whether the child is a girl. δ_p and λ_c represent prefecture and parental birth-cohort fixed effects, which absorb time-invariant local characteristics and cohort-specific trends, respectively. X_i' includes controls for parent and child characteristics, including parental education, occupation categories, dummy for co-residence with the child, child age, and child health status. The coefficient of interest, β_2, captures whether parents in prefectures with female Olympic medalists hold different educational expectations for daughters (relative to sons), compared to those in prefectures without female medalists.

Table 7 reports an asymmetry in how parents' educational expectations relate to female Olympic success. Panel A shows that in prefectures with female medalists, fathers expect their daughters to complete about 1.3 more years of schooling relative to sons, compared with fathers in prefectures without medalists. Decomposition by educational thresholds indicates that this effect is concentrated at the high school and college levels: fathers in exposed prefectures are 10 percentage points more likely to expect daughters (relative to sons) to complete high school, and 12.6 percentage points more likely to expect them to earn a college degree, with no significant difference at the junior-high level. These associations are specific to fathers; Panel B shows no corresponding patterns for mothers' expectations.

This parent-gender-specific pattern suggests that female Olympic success is associated with fathers forming higher beliefs about their daughters' educational potential, while mothers show no corresponding update in perceptions of their daughters' capabilities. This finding aligns with recent evidence that parents tend to overestimate gender gaps in competitiveness, with fathers' biased beliefs about daughters playing a particularly important role in shaping gender-differentiated parental choices (Yi, Heckman, Zhang, and Conti 2015; Bhaskar, Li, and Yi 2023; Tungodden and Willén 2023).

We next examine whether exposure to female Olympic medalists alters parents' educational investment patterns using expenditure data from the China Household Income Project (CHIP). Our analysis focuses on single-child households with children aged 6-18 across three survey waves spanning our study period (1988, 1995, 1999). The CHIP data provide rich household expenditure information, allowing us to construct two key spending categories: tuition fees and technical training expenditures.

Since the CHIP is a longitudinal survey spanning multiple waves, we can exploit the staggered timing of female medalists' emergence across prefectures. We estimate whether the daughter-son educational investment gap narrows after a hometown

Table 7. Effects on Parental Educational Expectations by Child Gender

	Dependent Variable: Parents' Educational Expectations for Children			
	Years of Schooling	Junior High	High School	College
Panel A: Fathers	(1)	(2)	(3)	(4)
Daughter × Medal Exposure	1.277**	0.0229	0.0998*	0.126**
	(0.640)	(0.0380)	(0.0557)	(0.0631)
Daughter	-0.742*	-0.00659	-0.0541	-0.0879*
	(0.392)	(0.0170)	(0.0340)	(0.0509)
N	640	640	640	640
R^2	0.803	0.587	0.700	0.780
Panel B: Mothers	(1)	(2)	(3)	(4)
Daughter × Medal Exposure	-0.539	-0.0194	-0.0318	-0.0158
	(0.501)	(0.0181)	(0.0387)	(0.0467)
Daughter	0.0290	0.00413	-0.00557	-0.0365
	(0.274)	(0.0111)	(0.0203)	(0.0277)
N	1,350	1,350	1,350	1,350
R^2	0.611	0.550	0.603	0.623
Controls	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y
BirthYear FE	Y	Y	Y	Y

Notes: This table examines how exposure to hometown female Olympic medalists is associated with parental educational expectations, separately for father (Panel A) and mother (Panel B) respondents, using data from the China Family Panel Studies (CFPS) 2010. The survey asks parents: "What is the highest level of education you hope your child will attain?" with response options ranging from illiterate to doctoral degree. The dependent variables are expected years of schooling (column (1)) and binary indicators for whether parents expect the child to complete junior high (column (2)), high school (column (3)), or college or above (column (4)). Medal Exposure is an indicator equal to one if the prefecture where the household is located had produced at least one female Olympic medalist prior to the survey year. All specifications include prefecture fixed effects and parental birth year fixed effects. Control variables include parental education, occupation categories, dummy for co-residence with the child, child age, and child health status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

female athlete wins an Olympic medal using the following specification:

$$\begin{aligned}\text{Expenditure}_{ipt} =& \beta_0 + \beta_1 \text{Daughter}_i + \beta_2 \text{Female Medal}_{pt} \\ &+ \beta_3(\text{Daughter}_i \times \text{Female Medal}_{pt}) + \gamma X_i + \delta_p + \theta_t + \varepsilon_{ipt}, \qquad (11)\end{aligned}$$

where Expenditure_{ipt} represents household expenditure in log for household i in prefecture p at survey year t; Daughter_i is an indicator for female child; Female Medal_{pt} indicates whether prefecture p had produced a female Olympic medalist by year t; X_{ip} includes controls for parental age, education, occupation, household size, and household income (in log); δ_p and θ_t represent prefecture and survey year fixed effects, respectively. The coefficient of interest, β_3, captures differential investment in

Table 8. Changes in Household Educational Expenditure

	Dependent Variable: Yearly Household Expenditure (RMB, log)			
	Tuition Fee	Training Fee	Housing Exp.	Healthcare Exp.
	(1)	(2)	(3)	(4)
Daughter × Female Medal	0.0452*	0.0175*	-0.0245	0.0131
	(0.0256)	(0.00991)	(0.0451)	(0.0312)
Daughter	-0.0621**	-0.0315**	0.0416	0.0151
	(0.0336)	(0.0167)	(0.0516)	(0.0215)
Female Medal	0.00554	0.00529	-0.0196	-0.0212
	(0.00661)	(0.00859)	(0.0467)	(0.0339)
Controls	Y	Y	Y	Y
Prefecture FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
N	32,513	32,513	32,513	32,513
R^2	0.125	0.081	0.347	0.215

Notes: This table examines the effect of exposure to hometown female Olympic medalists on parental educational investment using household expenditure data from the China Household Income Project (CHIP) surveys conducted in 1988, 1995, and 1999. The dependent variables are annual household expenditures in logarithm on tuition fees (column (1)), technical training fees (column (2)), housing (column (3)), and healthcare (column (4)). The sample is restricted to single-child households with children aged 6-18. All specifications include prefecture fixed effects and survey year fixed effects. Control variables include father's years of schooling, mother's years of schooling, parental hukou status, household size, and household income in logarithm. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

daughters' education when parents have been exposed to hometown female medalists.

Table 8 shows that the emergence of a female Olympic medalist is associated with a smaller gender gap in educational investment between daughters and sons. The pattern is consistent across two expenditure categories—tuition fees and training fees (columns (1) and (2)). These effects are economically significant: the estimated coefficients on the Daughter dummy are negative and significant in both columns, indicating that in the absence of female medalists, families invest considerably fewer educational resources in daughters than in sons. The emergence of female medalists narrows this gap, reducing the tuition disparity by 4.52 percent and the training expenditure gap by 1.75 percent.

To ensure our findings capture changes in educational investment rather than general household spending, we conduct placebo tests using non-education expenditures–housing and healthcare. As shown in columns (3) and (4), female medalists have no comparable effects on these categories, indicating that the observed shifts are specific to education. Overall, the results support our hypothesis that female Olympic success reshapes parental beliefs about daughters' potential and returns to education, reducing gender bias in household investment.

Transition and Dropout. These findings shed light on the transition patterns documented in Section 5.1, where female medalist exposure produces large effects at the junior and senior high school transitions but no significant effect at college entry. This heterogeneity reflects differences in parental influence across educational stages within China's highly selective system.

At the junior and senior high transitions, parents play a pivotal role in deciding whether to continue their children's schooling and bear the associated costs. These stages mark critical junctures where competitive tracking intensifies and families weigh the expected returns to further education. Female medalist exposure appears to shift these parental decisions, leading to higher female representation in senior high school. In contrast, college admission depends almost entirely on *Gaokao* performance, leaving little scope for parental influence. This explains why the conditional college enrollment gap among senior high graduates remains unchanged. Nevertheless, because more girls progress to senior high school, the overall likelihood that daughters attend college–accounting for earlier transitions–has risen relative to sons.

Academic versus Vocational Tracks. Beyond school continuation, parental decisions also shape the *track* of education that girls pursue. In China, students completing junior high school face a consequential sorting decision: enroll in an academic senior high school, which is the prerequisite for sitting the *Gaokao* and pursuing college admission, or enter a vocational secondary school leading directly to skilled-trade employment. Prior work interprets parents' choice between academic and vocational tracks as revealing their willingness to place children in competitive environments (Buser, Niederle, and Oosterbeek 2014; Reuben, Wiswall, and Zafar 2017).

Appendix Table A10 reports the results. Conditional on completing junior high school, exposure to female medalists raises girls' probability of progressing to an academic senior high school. The analogous effect at the tertiary level is null, consistent with *Gaokao* scores determining placement between regular and vocational colleges and leaving little room for parental discretion.

Norms and Attitudes. Beyond changes in parental beliefs and household investment, we also document broader shifts in societal attitudes toward gender. Using data from the Chinese General Social Survey (CGSS), we find that individuals exposed to hometown female medalists express more egalitarian views on gender roles and weaker adherence to traditional beliefs about women's domestic responsibilities (Appendix E).

Children's Self-Perceptions and Aspirations. The mechanism may also operate through girls' own self-perceptions: exposure to successful female athletes can shape how girls assess their abilities, competitiveness, and aspirations. We examine this possibility using the children's questionnaire from the 2010 wave of the CFPS, in which respondents aged

10–15 report their own attitudes rather than having parents answer on their behalf. We estimate a specification parallel to Equation (10), replacing parental expectations with children's self-reported attitudes (expected schooling, self-assessed academic performance and competence, effort, discipline, leadership, and career aspirations for competitive occupations) as the dependent variable.[21]

As reported in Appendix Table A8, we find no evidence that exposure shifts girls' self-reported attitudes—the interaction terms are statistically insignificant across all dimensions, contrasting sharply with the large and positive associations for fathers in Table 7. Young children's self-assessments may be less stable and precisely measured than parental expectations or actual investment behaviors, so we cannot fully exclude effects on girls' self-perceptions; nevertheless, the evidence suggests that belief updating among parents is likely the primary pathway through which female Olympic success influences educational outcomes.

We also examine whether female-medalist exposure shifts girls' *realized* career choices toward sport-related occupations using the population census. Across four occupational categories—professional athletes, sports agents, sports retail workers, and the combined sport-related sector—the coefficient is small and statistically insignificant (Appendix Table A9), consistent with the channel operating through parents rather than through girls' own aspirations or behavior.

Taken together, the evidence points to an asymmetric updating pattern: fathers (but not mothers) revise their beliefs about daughters' competitive potential; parental investment in daughters rises accordingly—both in education spending and in steering daughters toward academic rather than vocational tracks—while girls' own self-perceptions remain unchanged. Belief updating among parents, particularly fathers, thus appears to be the primary channel through which female Olympic success shapes educational outcomes. These findings offer a new perspective: beyond the financial constraints that prior work identifies as a key determinant of children's educational continuation (Cameron and Taber 2004), parental beliefs about competitiveness also shape children's educational progression. Together with the male-medalist null reported in Section 5.1, these patterns—unchanged girls' self-perceptions, no effect on sport-related career choices, and no effect on boys' educational outcomes—are difficult to reconcile with a same-gender role-model channel, in which individuals update primarily through identification with successful athletes of their own gender.

[21]Since the outcome of interest is children's attitudes, we include prefecture and child birth-cohort fixed effects. Control variables remain the same as in Equation (10).

7. Conclusion

Education is a crucial determinant of gender inequality, shaping gaps in employment, earnings, and fertility–particularly in developing countries (Goldin 2006; Duflo 2012; Jayachandran 2015). This paper examines whether and how female Olympic success reshapes beliefs about women's competitive capacity and advances gender equality in education. Using the Chinese context, we show that hometown female medalists significantly narrow gender gaps in years of schooling. To address potential endogeneity in medal timing, we employ a shift-share instrument based on the exogenous redistribution of medal opportunities following the Soviet Union's collapse, which reinforces the causal interpretation of our results.

We identify two complementary mechanisms underlying these effects. First, media framing matters: coverage emphasizing competitive-equality narratives–portraying female victories as evidence of women's capability in high-stakes competition–generates larger impacts than coverage centered on national pride. Second, exposure to female medalists systematically shifts fathers' beliefs about daughters' competitive potential, leading to higher educational expectations and greater household investment in daughters' schooling. These effects are specific to fathers and daughters, with no comparable responses from mothers or among sons. Together, the results demonstrate that publicized female competitive achievement can challenge entrenched gender stereotypes and yield measurable progress toward gender equality.

Our findings highlight two directions for future research. First, through what channels do these belief shifts diffuse beyond parent–child relationships? Understanding whether exposure shapes teachers' expectations or employers' behavior would clarify how counter-stereotypical exemplars catalyze broader social change. Second, the importance of media framing underscores the power of narrative in diffusing progressive values. Future work could assess whether similar communication strategies can promote inclusion and equality across other social domains.

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Online Appendix

A. Supplementary Figures

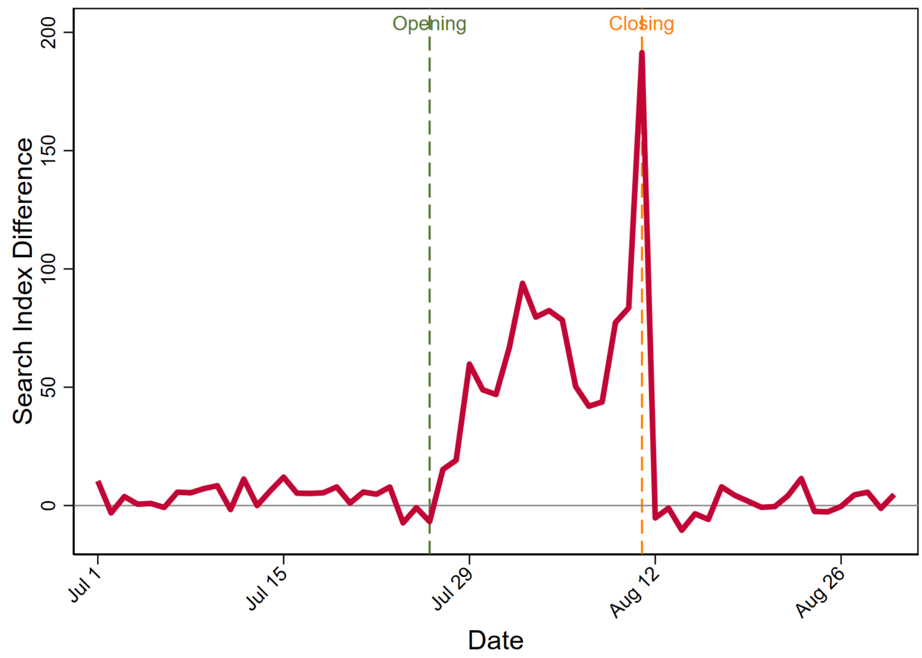


Figure A1. *Difference in Baidu Search Index for "Olympics" between medalist and non-medalist cities during the 2024 Paris Olympics. The Baidu Search Index measures daily keyword search volume on Baidu, China's dominant search engine, and serves as a proxy for local public attention analogous to Google Trends. For each city-day, the index reflects the relative frequency with which local users searched a given keyword, normalized to the national daily average. Medalist cities are defined as prefecture-level cities whose athletes won at least one medal at the 2024 Paris Olympics; non-medalist cities are those with no medal winners. Vertical dashed lines mark the opening (July 26) and closing (August 11) ceremonies. Before the opening ceremony, the difference hovers near zero, indicating comparable baseline interest across the two groups. During the Games, medalist cities exhibit substantially higher search volumes, peaking around the closing ceremony, before reverting to near zero in the aftermath. This pattern suggests that local athletic success generates a concentrated surge in community-level engagement with Olympic content, over and above the national interest shared by all cities.*

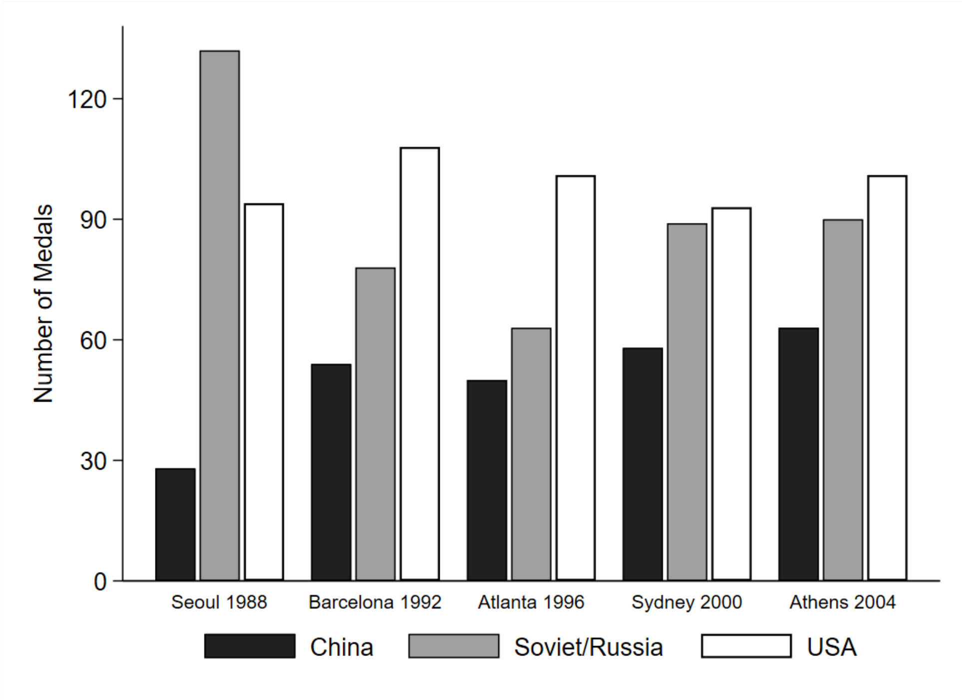


Figure A2. *Olympic Medal Statistics: China, Soviet Union/Russia, and USA (1988-2004). The figure shows the number of medals won by China, Soviet Union/Russia, and the USA across five Olympic Games from Seoul 1988 to Athens 2004. The data illustrates China's rise in Olympic performance following the Soviet collapse in 1991.*

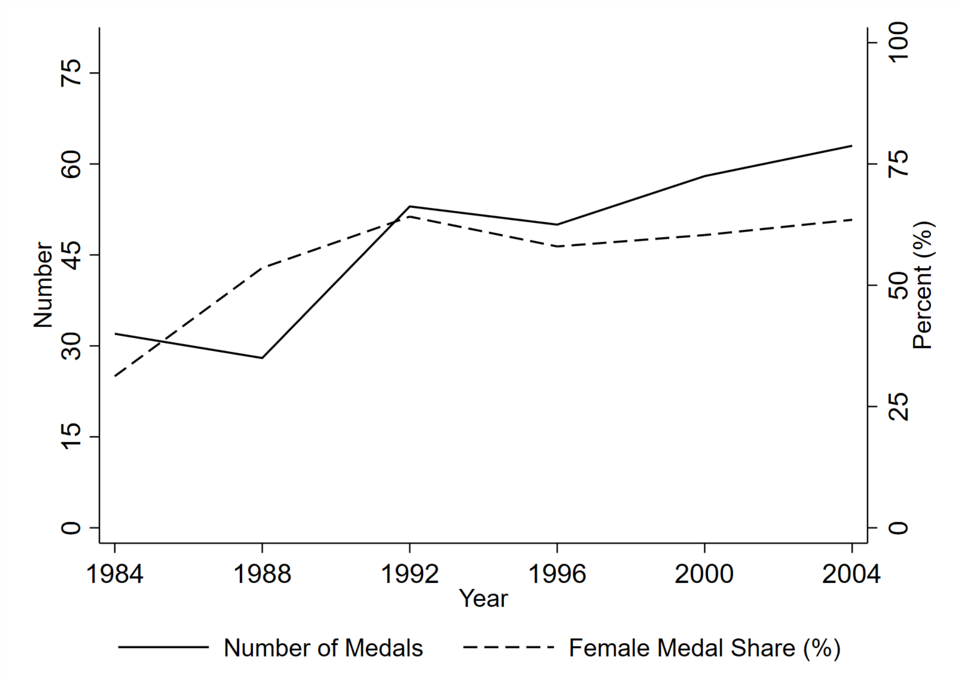


Figure A3. *Chinese Olympic Medals and Female Representation Over Time (1984-2004). This figure illustrates China's Olympic medal performance and female representation across six Summer Olympics from Los Angeles 1984 to Athens 2004. The solid line (left axis) shows the total number of medals won by Chinese athletes at each Olympic Games. The dashed line (right axis) represents the percentage of medals won by female athletes.*

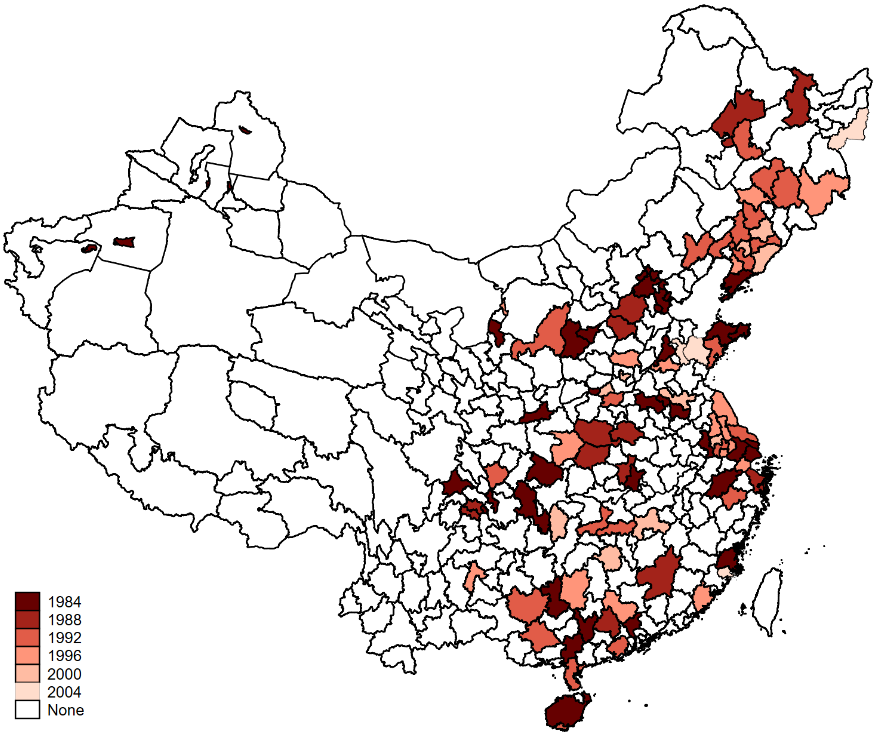


Figure A4. *Year of First Female Olympic Medalist by Hometown Prefecture (1984-2004). This map illustrates the geographic and temporal distribution of China's first female Olympic medalists across prefecture-level administrative units between 1984 and 2004.*

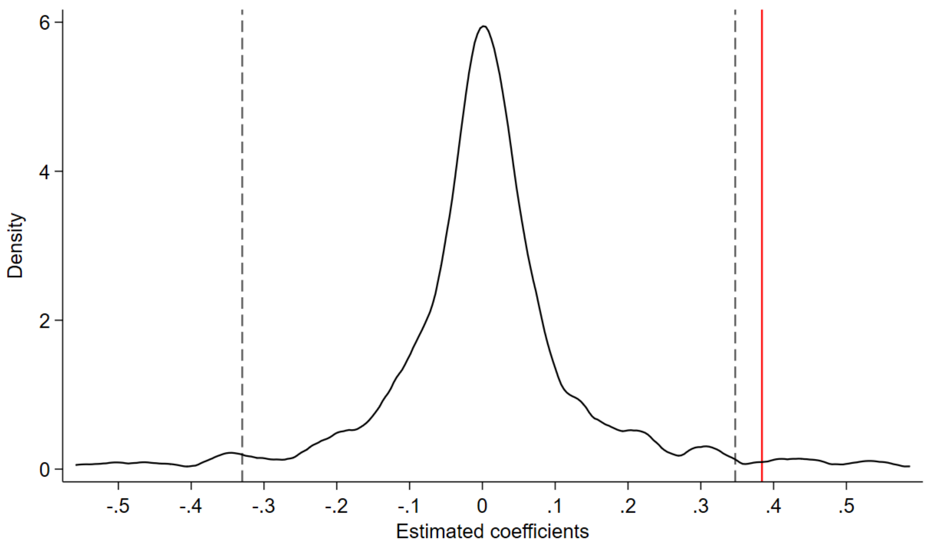


Figure A5. *Randomization Inference for the Recentered IV Estimate. This figure displays the distribution of second-stage IV coefficients obtained from 2,000 permutations of Soviet female participation rates across sports, used to construct the recentered instrument following Borusyak and Hull (2023). The solid curve shows the kernel density of the permutation-based estimates. The dashed vertical lines mark the 5% confidential interval of the distribution. The solid red line indicates our actual second-stage estimate of 0.384. The implied randomization inference p-value is 0.03.*

B. Supplementary Tables

Table A1. *Summary Statistics*

Variable	Obs	Mean	Std. dev.	Min	Max
Panel A: Demographic Characteristics					
Female	258,857	0.496	0.500	0	1
Birth year	258,857	1976.961	10.128	1960	1995
Han ethnic	258,857	0.947	0.223	0	1
Household hukou	258,857	0.966	0.181	0	1
Migrant	258,857	0.113	0.317	0	1
Panel B: Educational Attainment					
Years of schooling	258,857	10.646	3.368	0	19
Transition to primary school	258,857	0.960	0.196	0	1
Transition to junior high school	248,512	0.900	0.300	0	1
Transition to high school	223,585	0.497	0.500	0	1
Transition to college	111,043	0.523	0.499	0	1
Dropout during primary school	27,308	0.087	0.282	0	1
Dropout during junior high school	131,476	0.028	0.164	0	1
Dropout during high school	66,724	0.018	0.133	0	1
Dropout during college	58,536	0.010	0.100	0	1

Notes: This table presents summary statistics for the main analytical sample drawn from China's 2015 1% Population Mini-Census. The sample includes 258,857 individuals born between 1960 and 1995 from 91 prefectures that produced at least one female Olympic medalist between 1984 and 2004.

Table A2. *Top 10 Olympic Sport Events by Medal Count (1984-2004)*

Sport	Proportion (%)	Female Share (%)
Diving	13.57	47.37
Gymnastics	13.21	29.73
Shooting	12.14	38.24
Weightlifting	12.14	23.53
Table Tennis	11.79	57.58
Swimming	7.50	100.00
Badminton	7.50	80.95
Judo	5.00	100.00
Athletics	4.64	84.62
Fencing	2.50	57.14
Other	10.00	82.14

Notes: This table presents the top 10 Olympic sport events ranked by the number of medals won by Chinese athletes between 1984 and 2004, based on official records from the International Olympic Database."Other" includes all remaining sports not listed in the top 10.

Table A3. *Robustness Checks: Alternative Treatment Definitions*

	Dependent Variable: Years of Schooling		
	(1)	(2)	(3)
Female × Medal Cohort at 12	0.161***		
	(0.0530)		
Female × Medal Cohort at 15		0.192***	
		(0.0548)	
Female × Exposure Duration			0.0211***
			(0.00579)
Controls	Y	Y	Y
HomePref×BirthYear	Y	Y	Y
Female×Generation	Y	Y	Y
N	258,857	258,857	258,857
R^2	0.270	0.270	0.270

Notes: This table presents robustness checks using alternative definitions of treatment exposure. The dependent variable is years of schooling in all columns. Column (1) defines Medal Cohort as individuals aged 12 or younger when their prefecture first produced a female Olympic medalist. Column (2) uses age 15 as the cutoff. Column (3) uses a continuous treatment variable measuring the number of years individuals were exposed to female Olympic medalists before age 18. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A4. *Robustness Checks: Alternative Samples and Specifications*

	Dependent Variable: Years of Schooling									
	Excl. 1984 Medal Pref.	All Prefec- tures	All Age Co- horts	Incl. 2008 Beijing	Incl. Winter Olympics	Drop Mi- grants	Control Com- pul- sory Edu.	Control Gender- Pref. FE	Control GDP per Capita	Control Pri- mary Schools
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Female × Medal Cohort	0.161**	0.210***	0.200***	0.210***	0.200***	0.200***	0.186***	0.147***	0.200***	0.200***
	(0.0724)	(0.0394)	(0.0635)	(0.0572)	(0.0635)	(0.0652)	(0.0649)	(0.0446)	(0.0635)	(0.0635)
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Pref.×Birth Year FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Female×Pref. FE								Y		
Female×GDP p.c. (log)									Y	
Female×Primary Schools (log)										Y
N	165,413	768,528	359,409	298,380	258,857	229,593	258,857	258,857	258,857	258,857
R^2	0.212	0.285	0.337	0.273	0.270	0.278	0.270	0.272	0.270	0.270

Notes: This table presents robustness checks for the baseline fixed effects estimates. The dependent variable is years of schooling in all columns. Column (1) excludes prefectures that produced their first female Olympic medalist in 1984. Column (2) expands the sample to include all prefectures, including those that never produced a female Olympic medalist, while retaining the baseline birth cohorts. Column (3) expands the sample to include all age cohorts. Column (4) extends the medalist sample to include the 2008 Beijing Summer Olympics. Column (5) extends the medalist sample to include the Winter Olympics. Column (6) excludes individuals who migrated to other prefectures. Column (7) controls for prefecture-level compulsory education policy implementation interacted with birth cohort. Column (8) includes gender-by-prefecture fixed effects. Column (9) controls for the interaction between female and log GDP per capita. Column (10) controls for the interaction between female and the log number of primary schools. When prefecture-year data for GDP per capita or number of primary schools are missing, we impute using the corresponding provincial average for that year. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction. Control variables include ethnicity, hukou status, and migration status. Standard errors clustered at the prefecture level are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A5. *Geographic Spillover Effect of Female Olympic Medalists*

	Dependent Variable: Years of Schooling	
	Exposed Neighbors (Adjacent to Medal Cities)	Placebo Neighbors (Adjacent to Non-Medal Cities)
	(1)	(2)
Female × Medal Cohort	0.138	0.0111
	(0.0917)	(0.113)
Controls	Y	Y
HomePref×BirthYear	Y	Y
Female×Generation	Y	Y
Mean	9.73	9.81
N	70,264	102,986
R^2	0.237	0.207

Notes: This table examines the spillover effect of female Olympic medalist on gender disparities in educational attainment. The sample is restricted to non-medal cities, i.e., cities that did not themselves produce a female Olympic medalist. Column (1) reports the treated sample, consisting of non-medal cities geographically adjacent to a medal city and thus potentially exposed to spillover. Column (2) reports the placebo sample, consisting of non-medal cities whose adjacent cities are all non-medal cities and thus less likely exposed to spillover. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses are clustered at the prefecture level; * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

Table A6. *Heterogeneous Effects of Female Olympic Medalists by Sport Type*

	Dependent Variable: Years of Schooling			
	Competition Format		Athletic Attribute	
	Head-to-Head	Performance-Based	Strength-Oriented	Skill-Oriented
	(1)	(2)	(3)	(4)
Female × Medal Cohort	0.268**	0.160**	0.300**	0.149**
	(0.116)	(0.0723)	(0.122)	(0.0729)
Controls	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y
Mean	10.60	10.69	10.28	10.85
N	107,541	150,050	90,353	168,504
R^2	0.231	0.297	0.239	0.279

Notes: This table examines heterogeneity in the effect of exposure to hometown female Olympic medalists on gender disparities in years of schooling, distinguishing sports along two dimensions. Columns (1)–(2) split sports by *competition format*: *head-to-head* sports involve direct confrontation with opponents (e.g., Basketball), whereas *performance-based* sports involve competing against fixed standards or records (e.g., diving). Columns (3)–(4) split sports by *athletic attribute*: *strength-oriented* sports require substantial physical power (e.g., weightlifting), whereas *skill-oriented* sports rely primarily on technical proficiency (e.g., rhythmic gymnastics). In each pair, treatment is reassigned based on each prefecture's first female medalist within the relevant sport category. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses are clustered at the prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A7. *Soviet/Russian Female Athletes Participation by Sport (1988 vs 1996)*

	1988 Soviet		1996 Russia	
Sport	#Athletes	Proportion (%)	#Athletes	Proportion (%)
Athletics	44	8.35	47	6.37
Rowing	23	12.57	6	3.06
Handball	15	15.15	0	0.00
Basketball	12	14.46	11	8.40
Volleyball	12	14.29	12	9.60
Swimming	8	3.33	13	3.95
Gymnastics	6	7.14	7	7.14
Shooting	6	5.83	8	6.84
Canoeing	5	7.81	4	3.48
Fencing	5	7.81	6	7.14
Diving	4	11.11	4	7.55
Cycling	4	6.67	7	6.60
Synchronized Swimming	3	6.98	9	14.06
Table Tennis	3	6.67	2	2.60
Tennis	3	6.25	3	3.70
Archery	3	5.08	3	4.92
Rhythmic Gymnastics	2	5.41	8	9.64
Sailing	2	4.76	0	0.00
Equestrian	2	3.70	0	0.00
Judo	0	0.00	6	4.17
Hockey	0	0.00	0	0.00
Beach Volleyball	0	0.00	0	0.00
Softball	0	0.00	0	0.00
Badminton	0	0.00	2	2.35
Football	0	0.00	0	0.00
Average	8	7.67	6	4.46

Notes: This table shows the number and proportion of Soviet (1988) and Russian (1996) female athletes across Olympic sports. Proportions are calculated as the share of total female participants in each sport, excluding Chinese athletes. The data illustrates substantial cross-sport variation in Soviet female participation rates and the heterogeneous impact of political dissolution on competitive landscapes across different disciplines.

Table A8. Effects on Children's Self-Reported Attitudes

	Dependent Variable: Children's Self-reported Attitudes							
	Education	Achievement	Competence	Diligence	Compliance	Discipline	Leadership	Career
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Daughter × Medal Exposure	0.0893	-0.144	0.0272	0.00935	-0.244	0.0280	0.0272	-0.0607
	(0.252)	(0.115)	(0.111)	(0.656)	(0.232)	(0.407)	(0.111)	(0.0616)
Daughter	-0.0853	0.248***	0.199***	0.314	0.200	0.183	0.199***	0.0189
	(0.383)	(0.0697)	(0.0682)	(0.508)	(0.135)	(0.286)	(0.0682)	(0.0250)
Controls	Y	Y	Y	Y	Y	Y	Y	Y
HomePref FE	Y	Y	Y	Y	Y	Y	Y	Y
BirthYear FE	Y	Y	Y	Y	Y	Y	Y	Y
N	1,429	1,406	1,408	1,411	1,411	1,411	1,408	1,380
R^2	0.569	0.483	0.486	0.373	0.557	0.449	0.486	0.446

Notes: This table examines the association between exposure to hometown female Olympic medalists and children's self-reported attitudes using data from the China Family Panel Studies (CFPS) 2010. The dependent variables are: expected years of education (column (1)), self-assessment of academic performance on a 1-5 scale (column (2)), self-assessment of competence as a student on a 1-5 scale (column (3)), self-reported effort in studying on a 1-5 scale (column (4)), self-reported compliance with school rules on a 1-5 scale (column (5)), self-reported discipline in completing homework before playing on a 1-5 scale (column (6)), self-assessment of leadership capability on a 1-5 scale (column (7)), and preference for competitive careers—a binary indicator equal to 1 for careers involving high social evaluation, respect, income, or independence (column (8)). Medal Exposure is an indicator equal to one if the respondent's prefecture had produced at least one female Olympic medalist prior to the survey year. All specifications include prefecture fixed effects and child's birth year fixed effects. Control variables include parental education, occupation, survival status, coresidence with the child, child birth year, and child health status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A9. Female Medalist Effects on Sport-Related Occupation Choice

	Dependent Variable: Occupation Choice			
	Athlete	Sport Agent	Sport Retail	All Related
	(1)	(2)	(3)	(4)
Female × Medal Cohort	-0.000345	-0.0000448	-0.000122	-0.000512
	(0.000366)	(0.000480)	(0.000883)	(0.00109)
Controls	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y
Mean	0.0012	0.0018	0.0037	0.0067
N	258,857	258,857	258,857	258,857
R^2	0.012	0.015	0.020	0.018

Notes: This table examines whether exposure to hometown female Olympic medalists influences women's entry into sports-related occupations. The dependent variables are binary indicators for employment as professional athlete (column (1)), sports agent and related occupations (column (2)), sporting goods retail (column (3)), and all sports-related occupations combined (column (4)). All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A10. *Effect of Female Olympic Medal on Gender Disparities in Academic Track Choice*

	Dependent Variable: Education Track	
	Academic Senior High (1=Yes)	4-year College (1=Yes)
	(1)	(2)
Female × Medal Cohort	0.0140*	-0.00288
	(0.00838)	(0.00734)
Controls	Y	Y
HomePref×BirthYear	Y	Y
Female×Generation	Y	Y
Mean	0.428	0.272
N	223,585	111,035
R^2	0.140	0.134

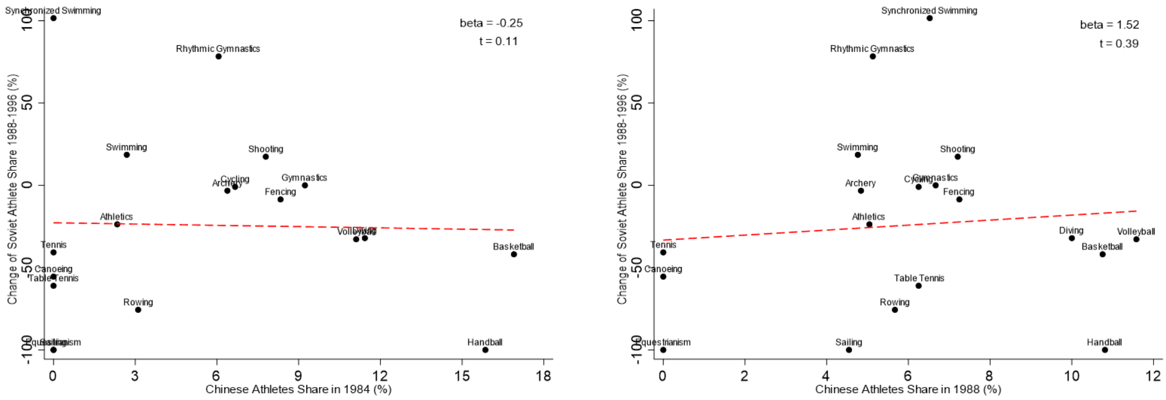
Notes: This table examines the effect of exposure to hometown female Olympic medalists on gender disparities in academic track choice at different educational stages. Column (1) restricts to individuals who completed junior high school; the dependent variable equals one if the individual progressed to an academic senior high school, and zero otherwise (i.e., enrolled in a vocational high school or did not continue schooling). Column (2) restricts to individuals who completed senior high school; the dependent variable equals one if the individual progressed to a regular (academic) college, and zero otherwise (i.e., enrolled in an associate/vocational college or did not continue schooling). All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses are clustered at the prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A11. *Treatment Cohorts: Year of First Female Olympic Medalist by Prefecture-Level City*

Year	**Cities**
1984	Beijing, Chengdu, Dalian, Fuzhou, Guangzhou, Hangzhou, Jinan, Liuzhou, Lüliang, Nanjing, Shanghai, Shangqiu, Suzhou (Jiangsu), Suzhou (Anhui), Taiyuan, Tianjin, Tonghua, Weihai, Wuhan, Wuzhou, Xi'an, Yantai, Yinchuan, Yulin (Guangxi)
1988	Baoding, Ganzhou, Nanyang, Neijiang, Ningbo, Qiqihar, Shijiazhuang, Xiaogan, Xiangyang, Yichun (Heilongjiang), Zhaoqing, Zigong, Zhumadian
1992	Anshan, Benxi, Changchun, Changsha, Changzhou, Chaoyang, Daqing, Haidong, Hechi, Jiangmen, Jilin, Jinhua, Jinzhou, Nanchong, Nanning, Nantong, Qingdao, Sanya, Shenyang, Tieling, Yiyang, Yulin (Shaanxi), Zhanjiang, Zhengzhou
1996	Guilin, Handan, Jiaxing, Liupanshui, Qingyuan, Shiyan, Siping, Tai'an, Taizhou, Wuhai, Wuxi, Yancheng, Yanbian†, Yangzhou, Yingkou, Zhangzhou
2000	Dandong, Fushun, Hebi, Hengyang, Jiaozuo, Liaoyang, Xiangxi‡, Xuzhou, Yichun (Jiangxi), Zhenjiang
2004	Jixi, Putian, Weifang, Zibo

Notes: This table presents treated prefecture-level cities grouped by treatment year, defined as the year in which the city produced its first female Olympic medalist.

C. Robustness Checks for the Shift-Share IV



(a) 1984 Los Angeles

(b) 1988 Seoul

Figure A6. *Soviet Decline and Pre-dissolution China Participation. These figures test whether the Soviet Union's decline in women's sports was systematically concentrated in sports where China posed a greater competitive threat. Each panel shows a binscatter plot examining the relationship between China's pre-dissolution athlete participation share (x-axis) and the subsequent change in Soviet athlete participation share (y-axis) across women's Olympic events. Panel (a) uses China's participation in the 1984 Los Angeles Olympics as the pre-dissolution measure, while Panel (b) uses the 1988 Seoul Olympics.*

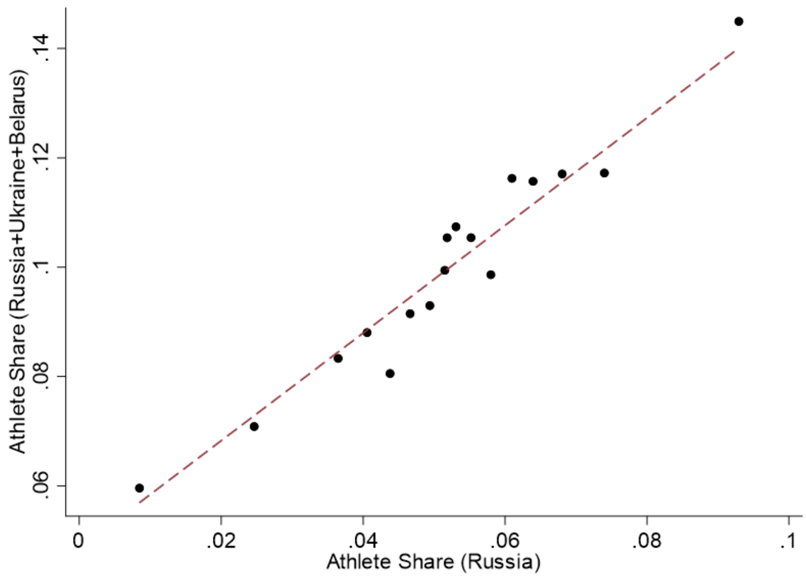


Figure A7. *Correlation between Russian and Combined Post-Soviet Athlete Participation Shares. This figure presents a binscatter plot examining the relationship between athlete participation shares across women's Olympic events for Russia alone (x-axis) versus the combined participation of Russia, Ukraine, and Belarus (y-axis) during 1996-2004. Each point represents the average participation share within bins of Russian participation, excluding Chinese athletes from the calculations. The analysis controls for sport event fixed effects. The strong positive correlation ($R^2 = 0.95$) demonstrates that Russian participation patterns closely mirror those of the broader post-Soviet region.*

Table A12. *Prefecture-level Characteristics before Soviet Dissolution and Predicted Medal Chance*

	Dependent Variable: Prefecture-level Characteristics in 1990						
	Sex Ratio *(%)*	GDP per Capita *(yuan)*	Primary Sector Share *(%)*	Secondary Sector Share *(%)*	Primary Schools *(per 1,000 pop.)*	Gyms *(per 1,000 pop.)*	Hospitals *(per 1,000 pop.)*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Predicted Medal 1992	-2.412	571.8	-4.616	-0.848	-10.76	-0.0278	2.753
	(3.812)	(632.0)	(4.971)	(4.819)	(7.130)	(0.0449)	(4.150)
Mean	0.52	2,310.52	24.55	47.57	46.38	0.24	25.28
N	91	84	84	84	84	83	84
R^2	0.001	0.010	0.010	0.000	0.027	0.005	0.005

Notes: This table examines whether the instrumental variable (predicted medal opportunities for the 1992 birth cohort) systematically predicts pre-determined prefecture-level characteristics. The dependent variables are measured before Soviet Dissolution: sex ratio (female to male) of cohorts born before 1991 (column (1)), GDP per capita in yuan in 1990 (column (2)), primary sector share of GDP in percentage points in 1990 (column (3)), secondary sector share of GDP in percentage points in 1990 (column (4)), number of primary schools per 1,000 residents in 1990 (column (5)), number of gymnasiums per 1,000 residents in 1990 (column (6)), and number of hospitals per 1,000 residents in 1990 (column (7)). Standard errors in parentheses; * p<0.1, ** p<0.05, *** p<0.01.

Table A13. Robustness Checks for Instrumental Variable Estimation

Control pre-1991 pref. char.	Instrumental Variable Estimation						
	Dependent Variable: Years of Schooling						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female × Medal Cohort	0.558***	0.652***	0.652***	0.652***	0.652***	0.669***	0.652***
	(0.173)	(0.193)	(0.193)	(0.193)	(0.193)	(0.198)	(0.193)
Controls	Y	Y	Y	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y	Y	Y	Y
Sex ratio×BirthYear	Y						
GDP per capital×BirthYear		Y					
1^{st} Industry Share×BirthYear			Y				
2^{nd} Industry Share×BirthYear				Y			
Primary School×BirthYear					Y		
Gym×BirthYear						Y	
Hospital×BirthYear							Y
F-statistics	36.171	36.171	36.171	36.171	36.171	34.817	36.171
N	258,857	234,556	234,556	234,556	234,556	232,539	234,556
R^2	0.003	0.002	0.002	0.002	0.002	0.002	0.002

Notes: This table presents robustness checks for the instrumental variable estimates. Columns (1)-(7) sequentially add interactions between pre-1991 prefecture characteristics and birth year fixed effects: column (1) controls for sex ratio × birth year, column (2) for GDP per capita × birth year, column (3) for primary sector share × birth year, column (4) for secondary sector share × birth year, column (5) for primary school density × birth year, column (6) for gymnasium density × birth year, and column (7) for hospital density × birth year. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A14. *Placebo Tests for Instrumental Variable Estimation*

	Dependent Variable: Years of Schooling			
	1960s Cohort	1970s Cohort	1980s Cohort	1960s-1980s Cohort
	(1)	(2)	(3)	(4)
Female × Medal Cohort	0.190	2.388	3.559	-0.0980
	(2.526)	(2.126)	(3.219)	(1.651)
Controls	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y
F-statistics	2.634	3.120	3.099	3.064
N	77,386	73,241	76,956	227,583
R^2	0.002	0.002	0.005	0.002

Notes: This table presents placebo tests examining whether the instrumental variable spuriously predicts educational outcomes for cohorts unaffected by the Soviet collapse. For each individual born in earlier cohorts (1960s-1980s), we assign predicted medal opportunities calculated for a corresponding later 1990s cohort from the same prefecture. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A15. *Bartik IV Regression: Female Olympic Medal Effects (Event Level)*

	Female Medal	
	1^{st} Stage	2^{nd} Stage
Dependent Variable	Female×Medal Cohort	Years of Schooling
	(1)	(2)
Female × Medal Cohort		0.437^{*}
		(0.221)
Female × Predicted Medal Cohort	0.296^{***}	
	(0.0725)	
Controls	Y	Y
HomePref×BirthYear	Y	Y
Female×Generation	Y	Y
F-statistics		16.629
N	258,857	258,857
R^2	0.888	0.003

Notes: This table reconstructs the instrumental variable using 214 detailed Olympic events rather than 29 broad sport categories. Column (1) shows the first-stage relationship between the event-level instrument and actual medal exposure, while column (2) presents the second-stage effect on years of schooling. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A16. *Defining Bartik IV Shift Using Athlete from Russia, Ukraine and Belarus*

	Sport-level Share		Event-level Share	
Dependent Variable	1^{st} Stage Female×Medal Cohort	2^{nd} Stage Years of Schooling	1^{st} Stage Female×Medal Cohort	2^{nd} Stage Years of Schooling
	(1)	(2)	(3)	(4)
Female × Medal Cohort		0.558***		0.437*
		(0.173)		(0.221)
Female × Predicted Medal Cohort	0.590***		0.296***	
	(0.0931)		(0.0725)	
Controls	Y	Y	Y	Y
HomePref×BirthYear	Y	Y	Y	Y
Female×Generation	Y	Y	Y	Y
F-statistics		40.174		16.629
N	258,857	258,857	258,857	258,857
R^2	0.892	0.003	0.888	0.003

Notes: This table examines robustness to alternative definitions of the "shift" variable in the Bartik-style instrument, using combined participation of Russia, Ukraine, and Belarus. Columns (1)-(2) use sport category to construct IV, while Columns (3)-(4) use more refined event category. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

Table A17. *Bartik IV Regression: Female Olympic Medal Effects (Re-centering)*

	Female Medal	
	1^{st} Stage	2^{nd} Stage
Dependent Variable	Female×Medal Cohort	Years of Schooling
	(1)	(2)
Female × Medal Cohort		0.384**
		(0.184)
Female × Predicted Medal Cohort	44.77***	
	(9.603)	
Controls	Y	Y
HomePref×BirthYear	Y	Y
Female×Generation	Y	Y
F-statistics		21.734
N	258,857	258,857
R^2	0.892	0.003

Notes: This table reconstructs the instrumental variable using a re-centering methodology. Column (1) shows the first-stage relationship between the re-centered instrument and actual medal exposure, while column (2) presents the second-stage effect on years of schooling. All specifications include prefecture-by-birth year fixed effects and the Female × Generation interaction term. Control variables include ethnicity, hukou type, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.

D. Text Analysis Methodology for Media Coverage Classification

To study media coverage as a mechanism for female medalists' effects on competitive equality, we employ text-based techniques to link media portrayals of athletes with their influence on local gender outcomes. In this section, we detail our methodology for systematically identifying and classifying thematic content in Olympic coverage, specifically how we define competitive-equality and national pride themes across different types of media reports to construct quantitative measures.

We develop comprehensive indices to capture competitive-equality and national pride themes by analyzing a large corpus from *People's Daily*, the official newspaper of the Chinese Communist Party's Central Committee. Given its central role in disseminating government policies and viewpoints, *People's Daily* provides a uniquely informative source for examining official narratives and policy-oriented language within Chinese media.

Our analysis begins with a dataset of 1,328,390 news articles published between 1946 and 2004. To focus specifically on content related to athletic achievements, we filter for articles mentioning Olympic medalists' names alongside terms such as "Olympics," "Sports," or "Competition." This process identifies 15,580 articles covering Olympic athletes, which we segment into 91,507 sentences for analysis.

A critical step in our methodology involves constructing well-defined training samples for classification through manual annotation. For the competitive-equality theme, we manually identify 500 sentences containing equal competitiveness discourse (e.g., phrases such as "hold up half the sky," "women are not inferior to men," and discussions of women's capabilities in traditionally male-dominated domains) and 5,000 sentences unrelated to this theme. Similarly, for the national pride theme, we manually annotate 500 sentences containing national pride narratives (e.g., phrases such as "glory for the motherland," "national honor," and discussions of athletic achievements elevating China's international standing) and 5,000 sentences unrelated to national pride. These manually labeled datasets serve as our training samples.

For classification, we employ a supervised learning approach by fine-tuning a pre-trained RoBERTa model, a transformer-based language model from the BERT family. We train separate binary classifiers–one for competitive-equality and one for national pride using an 80/20 train-validation split. Given the class imbalance in our training data (with non-related samples outnumbering related ones by approximately ten-to-one), we incorporate Focal Loss during training to assign higher weights to misclassified instances, effectively mitigating this imbalance. The Focal Loss function is particularly effective in addressing class imbalance by down-weighting easy examples and focusing learning on hard misclassified cases. The models demonstrate strong performance

Table A18. Keyword Sets for Classification

Competitive Equality Keywords	National Pride Keywords
Heroine	Strive for national glory
Outstanding woman	National honor
Resounding rose	National pride
March 8th Red Banner holder	Promote Chinese culture
Women of the new era	Live up to the motherland
Strong woman	National strength
Female general	National image
Exceptional woman	National spirit
Exemplary woman	National integrity
Female pioneer	National cohesion
Women's backbone	Repay the nation
Women's representative	Serve the nation
Women's army	Political responsibility
Advanced women's group	Historic breakthrough
Intellectual woman	
Female intellectual	
Gender equality	
Respect women	
Care for women	
Guarantee women's rights	
Protect women's interests	
Safeguard women	
Cultivate women	
Oppose arranged marriage	
Eliminate male preference	
Hold up for half sky	

Notes: This table presents the keyword sets used for constructing the training sample for classification of People's Daily articles into competitive-equality and national pride themes. The left column lists gender equality keywords and the right column lists national pride keywords. See Appendix D for detailed methodology.

on the validation set: the competitive-equality classifier achieves F1-scores of 0.96 for competitive-equality related sentences and 0.98 for non-competitive-equality sentences, while the national pride classifier achieves F1-scores of 0.96 for national pride related sentences and 0.99 for non-national pride sentences. These F1-scores correspond to over 95 percent classification accuracy on the validation set.

After training, the classifiers label every sentence in the full corpus of 91,507 sentences. An article is classified as competitive-equality-related or national pride-related if it contains at least one corresponding sentence. This procedure identifies 1,963 competitive equality-related sentences across 1,070 articles and 2,348 national pride-related sentences across 1,304 articles.

E. Complementary Evidence of Broader Attitudinal Change

In this section, we examine whether exposure to female Olympic success generates broader shifts in gender attitudes across society. We analyze data from the Chinese General Social Survey (CGSS) 2010, which contains comprehensive measures of gender attitudes across multiple dimensions. The survey asks respondents to indicate their agreement with six statements on a 5-point scale, ranging from "strongly disagree" to "strongly agree": (1) "Men should focus on career, women on family," (2) "Men are naturally stronger than women," (3) "Women are better off marrying well than achieving well," (4) "During economic downturns, women should be laid off first," (5) "Household duties should be shared equally between spouses," and (6) "It is equally good to have sons and daughters." These questions capture distinct dimensions of gender ideology, including beliefs about gender roles in the family and labor market, perceived gender differences in ability, and attitudes toward sons and daughters.

Based on these survey questions, we construct six indicators reflecting gender attitudes. For each measure, we define a dummy variable equal to 1 if the respondent chooses "somewhat agree" or "strongly agree," and 0 otherwise. For the first four statements, higher values indicate stronger endorsement of traditional gender norms; for the last two statements, higher values reflect more egalitarian attitudes. To examine how exposure to hometown female Olympic medalists affects these gender attitudes, we estimate fixed-effects specifications with each of the six attitude measures as the outcome variable.

Table A19 presents the results and suggests that female Olympic success shifts gender attitudes in a more egalitarian direction. Most notably, females exposed to female medalists are significantly less likely than males to agree that "men should focus on career, women on family" (column (1)), indicating weaker support for traditional gender specialization. At the same time, exposed females are significantly more likely to agree that "household duties should be shared equally between spouses" (column (5)), pointing to more egalitarian views of domestic responsibilities. For the other three traditional-attitude measures—natural strength, marriage versus achievement, and layoffs during downturns—the coefficients are also negative, although they are not statistically significant (columns (2)–(4)).

Finally, exposure to female medalists significantly increases agreement with the statement that "it is equally good to have sons and daughters" (column (6)). This finding suggests a weakening of son-biased attitudes rather than an active preference for daughters. Taken together, these results indicate that female Olympic success can reshape broader social attitudes about gender roles and the relative value of sons and daughters.

Table A19. *Changes in Gender Attitudes*

	Dependent Variable:					
	Career	Capacity	Marriage	Layoff	Housework	Daughter
	(1)	(2)	(3)	(4)	(5)	(6)
Female × Medal Cohort	-0.0715**	-0.0395	-0.0426	-0.0190	0.0825***	0.0418*
	(0.0278)	(0.0242)	(0.0329)	(0.0172)	(0.0248)	(0.0215)
Female	-0.0260	-0.0236	0.0517**	-0.00860	0.0546**	0.0180
	(0.0183)	(0.0200)	(0.0202)	(0.0114)	(0.0217)	(0.0109)
Medal Cohort	0.0572	0.0486	0.0256	0.00936	-0.0992***	-0.00513
	(0.0363)	(0.0403)	(0.0248)	(0.0214)	(0.0299)	(0.0225)
Controls	Y	Y	Y	Y	Y	Y
Prefecture	Y	Y	Y	Y	Y	Y
BirthYear	Y	Y	Y	Y	Y	Y
Mean	0.60	0.39	0.43	0.11	0.73	0.88
N	5,824	5,816	5,815	5,799	5,825	5,838
R^2	0.109	0.091	0.052	0.036	0.074	0.042

Notes: This table examines the effect of exposure to hometown female Olympic medalists on gender attitudes using data from the Chinese General Social Survey (CGSS) 2010. The dependent variables are binary indicators constructed from five-category responses—"strongly disagree," "somewhat disagree," "neither agree nor disagree," "somewhat agree," and "strongly agree"—and equal one if respondents choose "somewhat agree" or "strongly agree." Columns (1)–(6) correspond to agreement with the following statements: "Men should focus on career, women on family" (column (1)), "Men are naturally stronger than women" (column (2)), "Women are better off marrying well than achieving well" (column (3)), "During economic downturns, women should be laid off first" (column (4)), "Household duties should be shared equally between spouses" (column (5)), and "It is equally good to have sons and daughters" (column (6)). All specifications include hometown prefecture and birth year fixed effects. Control variables include ethnicity, hukou type, education, occupation categories, and migration status. Standard errors in parentheses clustered at prefecture level; * p<0.1, ** p<0.05, *** p<0.01.